

How Equal is Equal Opportunity?
Gender, Ethnic and Regional Disparities in Labor
Market Outcomes of Higher Education Policies in
Brazil

Master Thesis



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Acronyms

DiD Difference-in-Differences. 2, 19, I

ENADE Exame Nacional de Desempenho dos Estudantes. 10

ENEM Exame Nacional do Ensino Médio. 10

FIES Fundo de Financiamento Estudantil. 11, 21, 29

HEI Higher Education Institution. 1, 9, 18, 19

IBGE Instituto Brasileiro de Geografia e Estatística. 9, 16, 20

Lei de Cotas Lei nº 12.711/2012. 11

MEC Ministério da Educação. 10

PNAD Contínua Pesquisa Nacional por Amostra de Domicílios Contínua. 13

ProUni Programa Universidade para Todos. 1, 9, 12, 13, 15, 19, 52, I

RAIS Relação Anual de Informações Sociais. 1, 13, 14, I

SINAES Sistema Nacional de Avaliação da Educação Superior. 10

STF Supremo Tribunal Federal. 12

TWFE Two-Way Fixed Effects. 2, 19, 20, 29

1 Introduction

Higher education is widely regarded as an important instrument for promoting intergenerational mobility and reducing earnings inequality. Yet whether expansions in university access translate into broad and equitable labour-market gains remains an open empirical question, particularly in developing economies marked by strong regional and demographic disparities. Brazil, described by Bacha as *Belíndia*, a country that simultaneously resembles Belgium and India (Taylor & Bacha, 1976), offers a useful setting for empirical study: its labour market exhibits persistent wage gaps by ethnicity, gender, and region even after controlling for observable human capital (Arias, Yamada, & Tejerina, 2002; Telles, 2004; de Oliveira & da Mota Silveira Neto, 2021).

In 2005, the Brazilian federal government launched the Programa Universidade para Todos (ProUni), a large-scale scholarship programme that exchanges tax exemptions for private Higher Education Institution (HEI) participation in the provision of tuition-free university places for low-income students. By 2019, ProUni had allocated more than 3.5 million scholarships, and its institutional design included affirmative-action provisions reserving places for self-declared non-white applicants in proportion to their state-level population share (Ministério da Educação, 2024). The programme therefore constitutes both an expansion in higher-education access and a redistributive intervention aimed at historically disadvantaged groups.

Despite its scale and policy importance, the labour-market effects of ProUni remain insufficiently understood. Existing work has documented positive effects of affirmative action in elite public universities (Riehl, Machado, & Reyes, 2023) and estimated returns to education in Brazil using more conventional Mincerian approaches (Card, 1999; J. Mincer, 1974), but there is still limited causal evidence on how ProUni’s geographically uneven expansion affected local labour markets. In particular, little is known about whether the programme generated similar gains across demographic groups and across regions with very different higher-education capacity and labour-demand conditions.

This thesis addresses that gap by asking: *How does the expansion of ProUni scholarships affect formal-sector earnings, and do these effects vary systematically by gender, ethnicity, and geographic region?*

To answer this question, I construct a panel dataset linking three administrative sources, ProUni scholarship records, the *Relação Anual de Informações Sociais* (RAIS) employer–employee census, and Census population data, at the microregion level over the period 2005–2019. The empirical strategy builds a lagged cumulative scholarship-density measure intended to proxy the local expo-

sure of labour markets to ProUni-supported graduates, normalized by the youth population. Because the effects of educational expansion may be non-linear, I discretize this continuous exposure measure into dose strata and estimate heterogeneous treatment effects separately within those strata.

Identification relies on the heterogeneity-robust Difference-in-Differences (DiD) estimator of Callaway and Sant’Anna (2021), which is designed for settings with staggered treatment timing and potentially heterogeneous effects, and therefore avoids the well-known weighting problems of conventional Two-Way Fixed Effects (TWFE) regression (Goodman-Bacon, 2021; de Chaisemartin & D’Haultfœuille, 2020). The event-study implementation makes it possible to inspect pre-treatment dynamics transparently and to distinguish the primary causal objects of the analysis, within-stratum group-time average treatment effects, from more tentative cross-stratum comparisons. For the latter, the thesis follows the conceptual guidance of Callaway, Goodman-Bacon, and Sant’Anna (2024) and interprets dose contrasts cautiously.

The thesis makes three contributions. First, it provides reduced-form evidence on the local labour-market consequences of ProUni using a design that exploits both temporal and spatial variation in programme intensity; the doubly-robust specification yields a wage effect of approximately +3.0 per cent in moderately exposed microregions ($p < 0.10$), appreciably larger than the unconditional estimate. Second, distributional models by gender, ethnicity, and broad region reveal a non-monotone dose-response pattern consistent with diminishing marginal returns to local educational expansion in the most heavily exposed markets. Third, the naïve TWFE estimator produces a negative and marginally significant wage coefficient ($\hat{\beta} = -0.65\%$, $p = 0.083$) that masks the positive low-dose effect, providing a concrete empirical illustration of the forbidden-comparison bias documented by Goodman-Bacon (2021) and de Chaisemartin and D’Haultfœuille (2020).

Section 2 reviews the theoretical foundations and the empirical literature on returns to education in Brazil and internationally. Section 3 covers Brazil’s higher-education system, the design of ProUni, and concurrent policy interventions. Section 4 details the data sources, treatment construction, identification assumptions, and estimation strategy. Section 5 presents the main results, Section 6 interprets them, and Section 7 concludes.

2 Literature Review

2.1 Human Capital and Segmented Labor Market Theories

Returns to education and human capital have been subject to extensive empirical and theoretical investigation in economics. One of the first papers to formalise the interdependence between education and individual earnings was J. A. Mincer (1958), who later established that the logarithm of individual earnings can be expressed as a linear function of years of schooling and a quadratic in experience (J. Mincer, 1974). This earnings function became the dominant empirical framework in labour economics for half a century, yet it rests on strong assumptions, constant marginal returns to each year of schooling, separability of education and experience, that subsequent work has systematically questioned (Heckman, Lochner, & Todd, 2006).

Becker's ((1964)) theoretical framework predicts that the rate of return to college education should vary systematically across demographic groups once individual differences in financing costs and opportunity costs are accounted for. Empirically, he documented that estimated returns to schooling are positive even for historically disadvantaged groups, although the raw college wage premium was larger for white male cohorts, a pattern he attributed to unobserved heterogeneity in school quality and credit constraints rather than differences in underlying productive capacity (Becker, 1964, Ch. 4). This founding insight motivates the present study: if structural barriers rather than human capital deficits drive differential returns, then access-expanding policies like ProUni face an institutional ceiling that increases in education supply alone cannot lift.

In a comprehensive survey and meta-analysis of the causal returns to education literature, Card (1999) confirmed the positive direction of these results using causal inference methods and found that each additional year of schooling raises earnings by approximately 7–10%. Card (1999) further observes that policy-based instruments, such as school construction, proximity to colleges, and financial aid rules, typically yield Local Average Treatment Effect (LATE) estimates around 20% above OLS, reflecting the particularly high marginal returns earned by students who change their educational choices in response to such policies. He also highlights that “sheepskin” effects (Hungerford & Solon, 1987), the wage discontinuity at credential completion, may compound productivity gains with signalling premiums, and that these effects interact with ethnicity and gender (Belman & Heywood, 1991). While *human capital* theory interprets wage differentials as the result of individual investment outcomes in competitive markets, *segmented labour market* theory complements this view by emphasising institutional factors, hierarchies, and unequal access to the primary labour market (Reich, Gordon, &

Edwards, 1973; Doeringer & Piore, 1971; Leontaridi, 1998). The theory conceptualised in Reich et al. (1973) argues that the labour market is structurally divided into a primary sector, characterised by high wages, stability, and internal promotion ladders, and a secondary sector, characterised by low wages, instability, and limited returns to credentials. In this account, the allocation of workers to sectors is not solely determined by productivity but also by ethnicity, gender, and geographic origin, making segmentation a mechanism of structural reproduction rather than a neutral market outcome.

Leontaridi (1998) offers a synthetic assessment of the segmented labour market literature: while the theory captures important institutional realities, its explanatory power is weakened by inconsistent empirical definitions of what constitutes a “segment” and by difficulty in separating segmentation from compensating differentials. She recommends integrating segmented labour market insights with human capital and search models to produce more tractable empirical predictions. This methodological caution applies directly to the present thesis: the fact that ProUni operates through the private sector, structurally positioned in the secondary tier of Brazil’s dual university system, raises the question of whether its graduates gain access to the primary labour market or remain confined to the lower-return segment. The conclusions of both theories make it important to study the institutional mechanisms that drive the apparent heterogeneity in educational returns. If segmentation drives return differentials along demographic lines, then education policies solely aimed at expanding access may have a limited effect on reducing inequality or lifting people out of poverty. I therefore expect greater effects where policies open access to selective programmes and dense professional networks, thereby lowering ethnic and gender gaps, and smaller effects where regional labour demand limits the conversion of credentials into high-quality employment. Since most of the foundational studies cited above were conducted in the U.S. labour market, Section 2.3 surveys comparable evidence in international contexts, and Section 2.2 examines the specific Brazilian evidence.

2.2 Returns to Education in Brazil

The analysis of educational returns in Brazil must be contextualised within the country’s profound structural duality, termed *Belíndia* by the economist Edmar Bacha in the 1970s (Telles, 2004, p. 19). This blended word highlights the coexistence of a “small, growing industrial state such as Belgium” alongside a “great mass of poor stagnating at almost the level of rural India” (Taylor & Bacha, 1976). The entrenched economic segmentation justifies why educational policies must simultaneously address individual human capital deficits and the deep struc-

tural barriers that restrict social mobility between these two “countries.” Rather than treating economic agents as a homogeneous entity, this framework naturally leads to an analysis of the specific ethnic, gender, and regional market failures observed in Brazil.

Several contributions to Brazilian social science have employed methodologies in the tradition of J. Mincer (1974) to estimate educational returns with a pronounced focus on heterogeneity. Their collective message is that wage gaps in Brazil are stratified along three dimensions that interact with the mechanism of educational expansion evaluated in this thesis.

On ethnicity, Telles (2004, p. 171) documents income-distribution disparity comparable only to South Africa among major developing economies, identifying a glass-ceiling mechanism that almost entirely blocks non-white workers from ascending into the middle class. Arias et al. (2002) demonstrate an unexplained earnings gap of approximately 18% between non-white and white university graduates using historical pupil-teacher ratios as instruments for educational quality; Cornwell, Rivera, and Schmutte (2017) confirm its persistence in matched RAIS-PNAD data even after conditioning on observable worker and firm characteristics, providing direct evidence of labour-market discrimination beyond selection.

On gender, Bustelo et al. (2021) document that the Brazilian gender pay gap is substantially explained by occupational sorting at the point of college major selection: women disproportionately enter Health, Teaching, and Social Science programmes, which carry lower average wages in the formal sector, while men sort into Engineering, Technology, and Business with higher returns. Garcia, Nopo, and Salardi (2009) show using Ñopo’s non-parametric decomposition that this sorting effect, combined with within-occupation discrimination, sustained gender wage gaps of 20–25% throughout 1996–2006. A policy that expands access uniformly does not dissolve the downstream sorting and structural barriers that determine how credentials are priced in the labour market (Goldin, 2014).

On geography, de Oliveira and da Mota Silveira Neto (2021) document that average labour income in the metropolitan area of Fortaleza is more than 60% below the national average, and that women face a compounded disadvantage of lower regional wages and a larger unexplained within-region gender gap. This spatial heterogeneity is a central motivation for the microregion-level design: the “Brazil” side of Belíndia may be precisely where marginal returns to skilled labour are highest, yet where private HEI infrastructure is thinnest.

Policies can reshape these margins *unevenly*. Riehl et al. (2023) provide the most direct Brazilian evidence using a two-stage regression discontinuity design to study the direct and spillover effects of the 45% affirmative action quota at UERJ, an elite public university in Rio de Janeiro. They find a 14% increase in the

early-career earnings of marginally admitted non-white beneficiaries, attributable primarily to network access to high-paying firms rather than to human capital acquisition per se. Critically, however, these gains are decreasing in career progression, consistent with structural glass-ceiling barriers reasserting themselves as graduates age, and the spillover effect on top-ranked non-beneficiary enrollees in high-take-up majors is negative: early-career earnings for that group fall by 14%. Their results also confirm a spatial gradient in programme returns: gains are concentrated in the dense, productive labour market of São Paulo, while graduates in the Northeast capture substantially smaller premia. This spatial heterogeneity is a central motivation for the microregion-level design in this thesis.

2.3 International Evidence on Education Policies

The international evidence on large-scale education supply shocks offers two complementary lessons for the ProUni context. On one hand, Duflo’s ((2001)) analysis of Indonesia’s INPRES school construction programme, the closest institutional analogue to ProUni in terms of identification strategy, shows that regions receiving greater construction intensity experienced larger increases in schooling years and wages for exposed birth cohorts, with estimated annual returns of up to 10%. Duflo (2001) identifies these returns through the interaction of *spatial intensity* of the infrastructure shock with *cohort exposure*, the same two-dimensional identification logic employed in this thesis. A key finding is that the returns are concentrated among cohorts young enough to actually attend the new schools; older cohorts in the same districts show no effect, a pattern that motivates the age-cohort placebo design in Chapter 4.

A broader meta-analysis by Psacharopoulos and Patrinos (2004) across 139 countries confirms that returns to tertiary education are generally positive and often highest in developing economies where skilled labour is relatively scarce, consistent with the scarcity-versus-saturation hypothesis developed in this thesis.

Evidence from selective institutional settings reveals important distributional limits. Bagde, Epple, and Taylor (2016) show that India’s caste-based reserved seats altered college quality access and occupational sorting for socially disadvantaged groups, but that institutional quality acted as a binding constraint on the realised returns: beneficiaries assigned to lower-quality institutions captured smaller wage gains than those assigned to elite colleges. In the U.S. context, Bleemer (2022) finds that the elimination of affirmative action in California reduced minority enrolment at selective universities but also diminished downstream earnings, confirming that institutional selectivity is a key mediator of policy returns. Arcidiacono and Lovenheim (2020) synthesise this evidence and

argue that the quality-fit trade-off, whether beneficiaries are better served by a less selective institution where they perform well, or a more selective one where they struggle, depends critically on the local labour market and employer preferences.

Taken together, these studies establish three expectations that guide the empirical design of this thesis: (i) supply-side expansions generate positive wage returns for exposed cohorts when local labour demand is sufficient to absorb graduates; (ii) the magnitude of returns is highly heterogeneous across regions, demographic groups, and institutional quality tiers; and (iii) large-scale programmes may generate general equilibrium effects, including congestion and spillovers, that attenuate private returns in heavily exposed markets. The following chapters test each of these predictions in the Brazilian context.

3 Institutional Context

3.1 Higher Education in Brazil: A Structural Duality

Brazil's university system is among the most expansive in Latin America, characterized by a distinct public-private duality. According to the latest Census of Higher Education, the system comprises 2,580 institutions, of which more than 85% are private (INEP, 2024). While the federal and state public universities are tuition-free and historically elite, the private sector is highly heterogeneous, absorbing the vast majority of the demand for mass higher education.

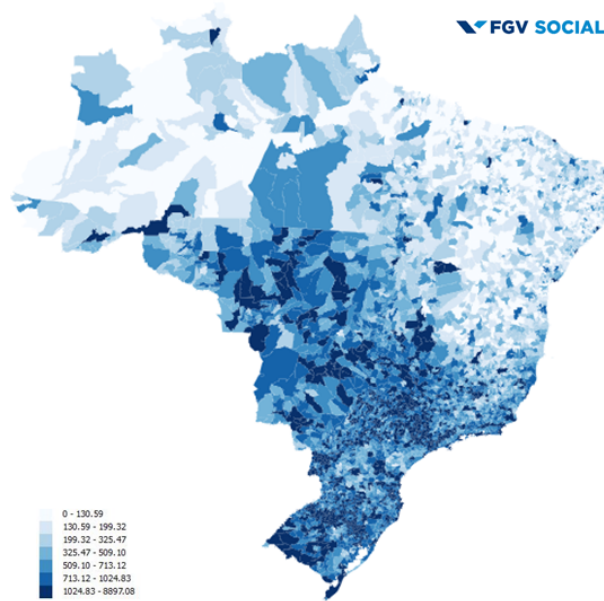


Figure 1: **The Geography of Inequality:** Average declared income by municipality (2020). The spatial concentration of wealth in the South/Southeast mirrors the historical concentration of university seats. Source: FGV Social (2020)

3.2 Historical and Sociological Foundations

Both ProUni's design and its unequal geographic distribution reflect deep features of Brazil's fiscal structure and racial sociology that took shape over the twentieth century.

3.2.1 The "Inverted Pyramid" of Public Financing

The necessity of a private-sector solution (ProUni) is rooted in the "inverted pyramid" of public financing (World Bank, 2002). The state allocates a disproportionate share of its budget to the tertiary level, which is free and dominated by the elite, while basic education remains underfunded. In the early 2000s,

public expenditure per student in higher education was approximately 13 times higher than in basic education (McCowan, 2007). This regressive fiscal structure constrained the expansion of the federal university system, necessitating a policy design like ProUni that exploited the private sector's marginal cost structure rather than exclusively expanding the capital-intensive public plant.

3.2.2 The Sociology of Race: From "Democracy" to Structural Racism

Unlike the United States, Brazil developed a unique ethnical order based on the ideology of "Racial Democracy", which held that extensive procreation between different ethnicities had created a fluid social structure free from rigid ethnical hatreds. However, as Telles (2004) documents, this ideology masked deep structural racism where economic inequality remained highly stratified.

A critical institutional challenge for policy is the complexity of racial classification. Brazilian classification is based on phenotype rather than ancestry. The IBGE uses five categories: *branca*, *preta*, *parda*, *amarela*, and *indígena*. However, the phenomenon of "money whitening", where individuals of higher socioeconomic status are more likely to perceive themselves as white, complicates the data (Telles, 2004). This structural reality justifies the aggregation of *Preta* and *Parda* into a single "non-white" category for the econometric analysis, as these groups share the structural disadvantage of the economically weaker sector.

3.3 The ProUni Program: Design and Formalia

The Programa Universidade para Todos (ProUni) was established by Law No. 11.096 in January 2005 to democratize access to higher education. By leveraging tax incentives, the program facilitated a massive expansion of the private sector's capacity.

As illustrated in Figure 2, the aggregate supply of scholarships grew exponentially following its implementation. Between 2005 and 2014, the annual intake of new beneficiaries increased from approximately 100,000 to over 300,000, accumulating a stock of human capital that represents the primary supply shock analyzed in this thesis.

3.3.1 Fiscal Mechanism: Tax Expenditures

ProUni is not a direct scholarship payment but a tax expenditure policy. Participating Higher Education Institutions (HEIs) receive exemptions from four federal corporate taxes, namely IRPJ, CSLL, PIS/Pasep, and COFINS, in exchange for offering scholarships (Campos, di Benedetto, & Pinheiro, 2019). To prevent the

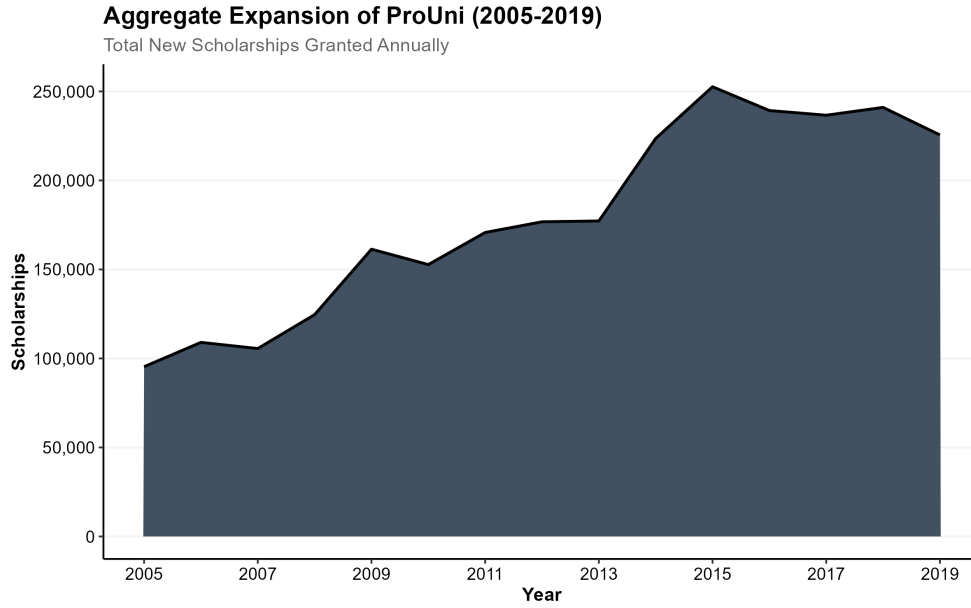


Figure 2: **Aggregate Expansion of ProUni Scholarships (2005–2019)**

Note: The figure displays the total number of new scholarships (full and partial) granted annually. The rapid expansion phase (2005–2014) constitutes the primary supply shock analyzed in this thesis. Source: Author's calculation using MEC micro-data.

proliferation of "diploma mills," the state linked these exemptions to quality regulation via the Sistema Nacional de Avaliação da Educação Superior (SINAES). Institutions that consistently underperform on the National Student Performance Exam (ENADE) face suspension from the program.

3.3.2 Eligibility and Selection Rules

The programme uses a strict means test and meritocratic selection based on the Exame Nacional do Ensino Médio (ENEM). Eligible applicants must have completed secondary education exclusively in public schools or as full scholarship students in private schools, must achieve at least an average ENEM score of 450 with a non-zero essay score, and must satisfy a household income test calibrated to the scholarship type: full (100%) scholarships require gross monthly household income per capita of at most 1.5 minimum wages, while partial (50%) scholarships require at most 3 minimum wages (Ministério da Educação, 2024).

3.3.3 Affirmative Action Mechanisms

ProUni mandates that institutions reserve a share of scholarships for self-declared *Pretos*, *Pardos*, and *Indígenas* proportional to their share in the respective state's population according to the latest Census (Ministério da Educação, 2024), effectively creating a voucher system for the non-white population to access the private

sector. Although there are no *de jure* gender quotas, the programme operates as a *de facto* gender policy: women constitute the majority of beneficiaries, driven by higher secondary completion rates and occupational sorting into Health and Education courses (Bustelo et al., 2021).

3.4 Concurrent Policies: FIES and the Quota Law

The evaluation of ProUni is complicated by two concurrent shocks that altered the supply of higher education. Figure 3 illustrates the temporal overlap of these interventions.

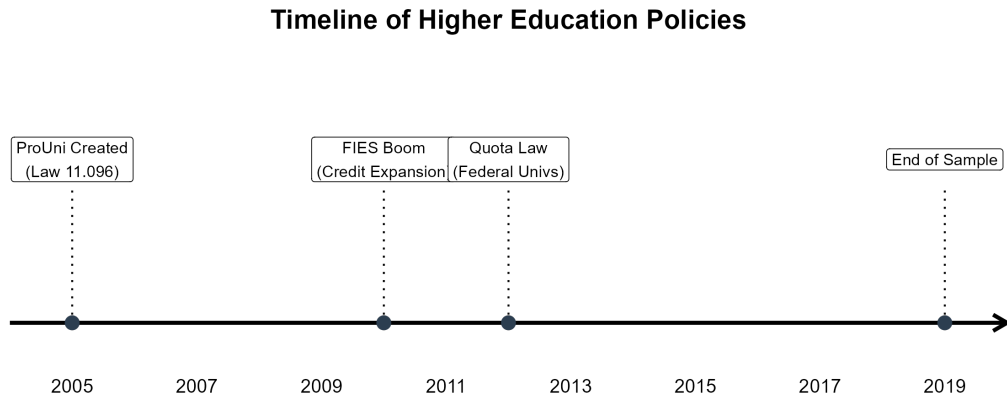


Figure 3: **Timeline of Higher Education Policies**

Note: The timeline highlights the three major interventions in the Brazilian higher education market. While FIES and the Quota Law overlap with the ProUni sample period, the identification strategy (Chapter 4) isolates the ProUni effect via the staggered adoption of scholarship density.

3.4.1 FIES: The Credit Expansion

Parallel to ProUni, the Student Financing Fund (FIES) provided subsidized loans. Between 2010 and 2014, FIES underwent an aggressive expansion with negative real interest rates, fueling a boom in private enrollments (Campos et al., 2019). Unlike ProUni, FIES students graduate with debt liabilities, theoretically altering their reservation wages.

3.4.2 The Federal Quota Law (Law 12.711/2012)

In 2012, the state intervened in the public sector with the Lei nº 12.711/2012 (Lei de Cotas). This law mandated that all federal universities reserve 50% of admissions for public school graduates, with sub-quotas for income and race (Riehl et al., 2023). This created a substitution effect: high-performing low-income students now choose between a free elite public seat (Quotas) and a

private scholarship (ProUni). This interaction motivates the inclusion of "Public University Expansion" controls in the empirical strategy.

3.5 Constitutional Foundations: Race, Quotas, and Enforcement

The shift from a colour-blind to a race-conscious state was settled by the Supreme Federal Court (STF) in ADPF 186, a 2012 landmark ruling in which the Court voted unanimously to uphold racial quotas. The prevailing opinion explicitly rejected formal equality in favour of material equality, holding that the state bears a duty of reparative justice toward groups historically marginalised by structural racism (Supremo Tribunal Federal, 2012).

The enforcement challenge, however, did not end with constitutionality. ProUni initially relied on self-declaration, but reports of fraud and the “money whitening” phenomenon, whereby higher-income individuals reclassify upward to access benefits, undermined targeting (Telles, 2004). The policy response was the creation of Bancas de Heteroidentificação (Heteroidentification Commissions), which verify phenotype rather than subjective identity, shifting the institutional definition of ethnicity to an administratively verifiable trait (Riehl et al., 2023). For the empirical analysis, this dynamic is a known source of measurement noise in the RAIS ethnicity variable: if reclassification is correlated with programme exposure, the measured effect on the non-white wage average conflates genuine wage gains with compositional change in who is counted as non-white.

4 Data and Empirical Strategy

4.1 Introduction

This chapter details the identification strategy employed to evaluate the distributional labor market consequences of the Programa Universidade para Todos (ProUni). The central empirical challenge is to isolate the causal effect of a large-scale, staggered supply shock on local earnings from confounding factors such as secular growth trends or pre-existing regional divergence, especially in the context of Brazil’s economic dualism.

Following the seminal framework of Duflo (2001), I exploit two distinct dimensions of variation: the *spatial intensity* of the program (the density of scholarships allocated to specific microregions) and the *temporal exposure* of different cohorts (determined by the approximate timing of entry into the labor force). This interaction allows for a precise “dosage” design: rather than asking merely whether

ProUni affected wages, this thesis investigates how the magnitude of the supply shock shapes labor market outcomes on both the extensive and intensive margins.

Departing from the parametric linear interaction models common in earlier literature, I adopt the *forward-engineering* approach recommended by Baker, Callaway, Cunningham, Goodman-Bacon, and Sant’Anna (2025): I first define the target causal parameter (group-time average treatment effects within dose strata), state identification assumptions explicitly, and then estimate each 2×2 building block separately before aggregating transparently.¹ Specifically, I construct a continuous, lagged measure of ProUni exposure (a scholarship-density “dose”), partially aggregate this continuous dose into interpretable exposure strata, and estimate group-time average treatment effects using the `did` package of Callaway and Sant’Anna (2021), aggregating them into dynamic event studies. This approach relaxes the strict assumptions of constant treatment effects and linearity, allowing for the explicit detection of non-linear mechanisms such as market saturation and scarcity rents.

The chapter proceeds as follows. Section 4.2 describes the three administrative data sources and their harmonization. Section 4.3 constructs the treatment variable. Section 4.4 presents the identification assumptions and estimation equations. Section 4.5 details the validation protocol.

4.2 Data Sources and Processing

The analysis integrates three primary administrative data sources, harmonized via the `basedosdados` infrastructure (Cavalcante, Herszenhut, & Dornelles, 2022) to ensure consistent geographic identifiers and variable definitions across the 2005–2019 period. This package sources data directly from the API of Base dos Dados, a non-profit open data platform that centralizes and harmonizes hundreds of public Brazilian datasets (Base dos Dados, 2026; Dahis, Carabetta, Scovino, Israel, & Oliveira, 2022). The data visualization in this thesis is facilitated by the `geobr` package developed by Pereira and Goncalves (2025).

4.2.1 Labor Market Outcomes: RAIS

Data description. The primary source of the dependent variables is *Relação Anual de Informações Sociais* (RAIS), a matched employer–employee administrative census of Brazil’s formal labor market created by decree in 1975. Unlike household surveys (for example PNAD Contínua), which are subject to sampling error and self-reporting bias, RAIS provides near-universe coverage of formal

¹This follows the methodological framework of Callaway and Sant’Anna (2021) for staggered binary treatments and Callaway et al. (2024) for continuous treatments.

earnings with geographic detail down to the municipal level (Ministério do Trabalho e Emprego (MTE), 2024; Instituto Brasileiro de Geografia e Estatística (IBGE), 2024; Dix-Carneiro & Kovak, 2017). Employer reporting is mandatory, and non-compliance is punishable by fines, underpinning the high coverage of formal employment in the data (Engbom, Gonzaga, Moser, & Olivieri, 2022).

RAIS is organized around the employment relationship (*vínculo*): each row links a worker to an establishment in a given year and reports both worker characteristics (for example age, ethnicity/phenotype, gender) and job characteristics (for example remuneration) (Ministério do Trabalho e Emprego (MTE), 2024).

Sample construction. To focus on the demographic directly exposed to the policy, I restrict the outcome sample to individuals aged 22 to 35 years. This age band captures the first generation of ProUni graduates entering the workforce (typically aged 22–24 at graduation) and their early career trajectory, while excluding older cohorts whose wage dynamics are primarily driven by seniority or tenure and would introduce noise into the estimation of returns to schooling (Card, 1999; Angrist & Pischke, 2009). The panel strictly ends in 2019 to avoid the structural break caused by the COVID-19 pandemic, ensuring that the parallel trends assumption is not violated by asymmetric economic shocks in 2020.

Age-Cohort Decomposition. Following Duflo (2001), who identified the returns to Indonesia’s school construction program by comparing young cohorts (who could use the new schools) against older cohorts (who could not) within the same district, I decompose the main 22–35 sample into two age cohorts. The exposed cohort (ages 22–28) consists of workers who could plausibly be ProUni graduates, having entered university at the standard age (18–20) and graduated within 4–6 years; the control cohort (ages 29–35) consists of workers too old to have been ProUni beneficiaries at standard university-going age during the treatment period. Both cohorts are downloaded separately from RAIS using the same SQL structure, sample restrictions, and geographic aggregation as the main extract.

Main outcomes. The main dependent variable is the average of log monthly remuneration (in minimum wages) of formal workers aged 22–35. Specifically, I first compute $\ln(w_i)$ at the individual worker level within each municipality–year–demographic cell, then aggregate to the microregion–year cell by taking the population-weighted mean across municipalities within the microregion (Section 4.2.3):

$$\overline{\ln w_{r,t}} = \frac{\sum_{m \in r} N_{m,t} \cdot \overline{\ln w_{m,t}}}{\sum_{m \in r} N_{m,t}}, \quad (1)$$

where $\overline{\ln w_{m,t}}$ is the average of individual log wages in municipality m at time t and $N_{m,t}$ is the number of formal workers aged 22–35 in that municipality–year cell.² The main estimation sample comprises non-white workers (RAIS race codes 1, 4, and 8: Indígena, Preta, and Parda), consistent with ProUni’s affirmative action mandate targeting historically disadvantaged populations. White workers (code 2, Branca) are analysed separately in the heterogeneity subsample.

In addition, I construct an extensive-margin outcome: the log total number of formal *vínculos* of individuals aged 22–35 in each microregion–year cell, $\ln N_{r,t}$, distinguishing effects operating through wages from those through employment.

Summary statistics for both outcomes, stratified by dose group, are reported in Table 1 in Section 5.

4.2.2 Treatment Intensity: ProUni

Treatment data is derived from the administrative records of the ProUni programme (`basedosdados.br_mec_prouni.microdados`), recording the beneficiary’s home municipality (`id_municipio`), institution identifier (`id_ies`), year, scholarship type (*integral* or *parcial*), and demographic characteristics of every scholarship granted between 2005 and 2019.

Three aspects of this treatment construction merit comment. First, scholarship counts are attributed to the beneficiary’s home microregion rather than the HEI’s location. This demand-side assignment is appropriate for a labour-market study: the theoretical channel runs through the treated worker re-entering their local labour market after graduation, and RAIS workplace municipality correlates strongly with residence municipality for Brazilian formal-sector workers.³ Second, to measure intensity rather than volume, scholarship counts are normalised by the local youth population (aged 18–24) obtained from the 2010 Demographic Census, so that the treatment variable reflects the saturation of the potential skilled labour force. Third, the baseline dose counts all scholarship grants equally (*integral* and *parcial*); as a robustness check, I compute a full-scholarship-equivalent

²This construction follows the standard Mincerian tradition in labor economics. The alternative, taking the log of the arithmetic mean wage, $\ln(\bar{w}_{r,t})$, conflates wage level effects with within-unit distributional shifts via Jensen’s inequality: $\ln(E[w]) \geq E[\ln(w)]$, with the gap driven by within-municipality wage variance. If the treatment changes the within-unit wage distribution, the log-of-average estimand captures both level and inequality effects, complicating interpretation. The average-of-logs measures changes in the geometric mean wage, is less sensitive to outliers, and corresponds to percentage changes in individual welfare under concave utility (Card, 1999).

³An HEI-location (supply-side) dose, obtained by joining `id_ies` to the `basedosdados.br_bd_diretorios_brasil.instituicao_ensino_superior` directory, is used as a robustness check in Section 4.5. The two doses are highly correlated ($r > 0.90$ in the 2019 cross-section), confirming that private HEIs are predominantly attended by local residents.

(FSE) dose that weights integral grants at 1.0 and parcial grants at 0.5 (reflecting their 100 per cent and 50 per cent tuition coverage), and the FSE and raw-count doses are highly correlated ($r > 0.99$) and yield virtually identical dose-bin classifications.

Table ?? reports the demographic composition of scholarship recipients over the programme period. Consistently with the programme’s affirmative-action design (Ministério da Educação, 2024), a majority of grants accrue to female and non-white (Parda, Preta, Indígena) beneficiaries.

4.2.3 Unit of Analysis: The Microregion

While Brazil is administratively divided into 5,570 municipalities, this analysis aggregates data to the Microregion level ($N = 558$), an official regional division above municipalities that the IBGE established in 1990 according to (a) the spatial structure of industrial and agricultural production and (b) the flow of goods, capital, and information (Instituto Brasileiro de Geografia e Estatística (IBGE), 2022).⁴

This aggregation is necessary to address the “spatial mismatch” present in municipality-level data. In integrated economic zones, an individual’s residential location (where the scholarship is theoretically assigned) often differs from their workplace location (where the earnings outcome is measured) due to commuting patterns. As noted in the literature on Brazilian local labor markets (Dix-Carneiro & Kovak, 2017), the Microregion can be regarded as a self-contained commuting zone in which labor supply and demand equilibrate. This approach minimizes measurement error from cross-municipal commuting while maintaining 558 clusters, well above the threshold required for valid cluster-robust inference (Cameron & Miller, 2015).⁵

In practice, outcome variables are constructed from municipality-year RAIS aggregates and then spatially aggregated to the microregion-year level via the mapping `id_municipio` $\rightarrow$ `id_microrregiao`, using population weights to ensure that the microregion measure reflects the size composition of its constituent municipalities.

⁴IBGE’s meso/micro regionalization has been superseded by the Immediate and Intermediate Geographic Regions in 2017, but microregions remain a standard unit in applied work with Brazilian administrative data that reduces municipality-level commuting measurement error (Instituto Brasileiro de Geografia e Estatística (IBGE), 2022).

⁵The “few clusters” problem identified by Cameron, Gelbach, and Miller (2008) arises primarily with 5–30 clusters. With 558 clusters, standard asymptotic cluster-robust standard errors are fully reliable.

4.3 Defining the Treatment: The “Dose” of Human Capital

A central hypothesis of this thesis is that educational supply shocks exhibit non-linearities. A linear continuous treatment specification, for example “effect per 1 additional scholarship”, assumes that the 100th scholarship has the same marginal effect as the 10,000th. This is unlikely to hold if local markets face absorptive capacity constraints. To capture potential non-linearities, I adopt a discretized continuous treatment approach following Callaway et al. (2024).

4.3.1 The Continuous Dose Variable ($D_{r,t}$)

For each microregion r and year t , I construct the cumulative scholarship density variable $D_{r,t}$:

$$D_{r,t} = \frac{\sum_{k=2005}^{t-4} \text{Scholarships}_{r,k}}{\text{Youth Population}_{r,2010}} \times 1000. \quad (2)$$

The numerator is lagged by four years. Following the logic in Duflo (2001), an educational investment in year t only becomes a labor supply shock when beneficiaries plausibly complete their degree and enter the workforce. Accordingly, I interpret the lagged dose as a proxy for the local inflow of ProUni-educated workers, noting that the numerator captures *scholarship grants* (not observed degree completions) and therefore represents an exposure measure rather than a literal count of graduates.

The denominator uses the 2010 Census youth population as a time-invariant normalizer, ensuring that the dose variable is not contaminated by demographic changes that may themselves be affected by the treatment. While the ideal normalizer would be the 2000 Demographic Census to guarantee strict pre-treatment exogeneity, the harmonisation of microregional boundaries between 2000 and 2010 presents severe data linkage challenges. The use of the 2010 youth population relies on the assumption that inter-microregional migration specifically induced by ProUni between 2005 and 2010 was small relative to baseline demographic stocks. Given the geographic vastness of microregions, cross-border migration for private (non-elite) tertiary education is institutionally less common than local enrollment, mitigating this endogeneity concern.

4.3.2 Discretization

Following the methodological recommendations of Callaway et al. (2024) for handling continuous treatments with high dimensionality, I partially aggregate the continuous exposure measure into a small number of empirically meaningful dose strata, yielding readable “dose-aware” event-study plots that retain heterogeneity across the treatment distribution.

Bin definition. I use the cross-sectional distribution of $D_{r,2019}$ as the baseline way to classify microregions into dose strata. This choice reflects the full realized treatment intensity of the program and maximizes cross-sectional variation for distinguishing low- and high-exposure markets. At the same time, because the classification is based on realized end-of-sample exposure, it should be viewed as an empirical partition rather than a purely pre-treatment object.⁶

This choice is motivated by the institutional structure of ProUni’s allocation mechanism. Scholarship supply is determined by *national* budget decisions and the presence of private HEIs willing to participate in exchange for tax exemptions. MEC does not mechanically direct more scholarships to microregions because their current wages are rising faster. Nevertheless, the 2019 distribution is strongly correlated with pre-existing private HEI infrastructure, urban scale, and regional economic structure. For that reason, all cross-dose comparisons in the thesis are interpreted cautiously: the within-stratum ATT estimates are the primary causal objects, while contrasts between dose strata are presented as suggestive unless stronger assumptions are invoked.

The quantile cutoffs from the 2019 distribution define four groups. The control group comprises microregions whose lagged dose has not yet exceeded the 20th percentile threshold τ_{20} of the 2019 distribution ($D_{r,t} \leq \tau_{20}$); these regions received negligible exposure and serve as the counterfactual for “no effective treatment.” The Low-Dose (scarcity) stratum collects microregions with 2019 density between the 20th and 50th percentiles ($D_{r,2019} \in (\tau_{20}, \tau_{50}]$), corresponding to a moderate supply shock and providing the empirical test of whether returns are preserved under relatively scarce expansion. The Mid-Dose stratum collects microregions with 2019 density between the 50th and 75th percentiles ($D_{r,2019} \in (\tau_{50}, \tau_{75}]$); these regions experienced an intermediate supply shock, with median treatment year 2013, earlier than the Low-Dose median of 2017 but later than the High-Dose median of 2012. The High-Dose (saturation) stratum collects microregions with 2019 density above the 75th percentile ($D_{r,2019} > \tau_{75}$), corresponding to a large supply shock and providing the test of whether returns are attenuated in heavily exposed markets.

This three-stratum partition traces the theoretically motivated transition from scarcity, through intermediate exposure, to saturation, while maintaining adequate sample sizes within each bin for reliable inference. Robustness to alternative

⁶An alternative approach would define bins from the *initial treatment year* distribution D_{r,G_r} , avoiding any potential look-ahead concern. In practice, the early-period distribution is extremely compressed (all microregions have near-zero density in 2009 when the lag first activates), yielding noisy bin assignments that attenuate treatment contrasts. I therefore retain the 2019-based partition as the baseline but report alternative cutoff and timing schemes as sensitivity checks in Section 4.5.

cutoff choices (τ_{25}/τ_{75} and τ_{33}/τ_{67}), and to replacing raw scholarship counts with full-scholarship-equivalent (FSE) weights (integral = 1.0, parcial = 0.5), is assessed in Section 4.5. The FSE and raw dose measures yield a bin-reclassification rate below 5 per cent, confirming that programme-level results do not depend on the within-year scholarship-type aggregation.

Treatment-timing variable. The treatment-timing variable G_d^r used in the staggered event-study design is defined as the *first year* in which microregion r 's lagged density $D_{r,t}$ crosses the low-exposure threshold τ_{20} :

$$G_d^r = \min\{t \in [2009, 2019] : D_{r,t} > \tau_{20}\}, \quad (3)$$

with $G_d^r = \infty$ for any microregion whose density never exceeds τ_{20} . This ensures a consistent event-time origin $e = t - G_d^r$ for both the Low-Dose and High-Dose subsamples.

Comparison group. Since all 558 microregions eventually accumulate positive scholarship exposure over the 2005–2019 period, no microregion satisfies $D_{r,t} = 0$ permanently. The identification strategy therefore relies on not-yet-treated microregions as the comparison group, as explicitly sanctioned for staggered designs without a never-treated group by Callaway and Sant'Anna (2021) (Section 5) and Callaway et al. (2024). Formally, the comparison group for cohort $G_d^r = g$ at time t is:

$$C_t = \{r : G_d^r > t\}, \quad (4)$$

consisting of all microregions that have not yet crossed the dose threshold τ_{20} by period t . When estimating $ATT_d(g, t)$ for the Low-Dose subsample, units in the High-Dose bin are excluded from both the treated and the comparison sets, so that each dose-specific event study is internally coherent. This is implemented in R via `did::att_gt()` with `control_group = "notyettreated"`.

4.4 Empirical Strategy: Heterogeneous DiD

Identifying the causal effect of ProUni faces two central econometric challenges: *staggered treatment adoption* and the *likelihood of heterogeneous effects across cohorts and over time*. These challenges are particularly acute for ProUni because the program rolled out first in microregions with pre-existing private HEI infrastructure, making early adopters structurally different from late adopters. In such settings, the standard Two-Way Fixed Effects (TWFE) estimator compares treated units not only to untreated units but also to already-treated units,

so-called “forbidden comparisons” (Goodman-Bacon, 2021). If treatment effects evolve over time, these comparisons receive negative weights that can severely distort estimates, potentially flipping the sign of the coefficient (de Chaisemartin & D’Haultfœuille, 2020; Goodman-Bacon, 2021). I therefore employ the non-parametric estimator proposed by Callaway and Sant’Anna (2021) as the primary specification, benchmarking against TWFE as a diagnostic (Section 4.5.8).

4.4.1 Identification Assumptions

The causal interpretation of the group-time average treatment effects estimated in this framework rests on four assumptions, stated below.

Assumption A1 (Parallel Trends). In the absence of treatment, average potential outcomes in treated and comparison microregions would have followed parallel paths. Since no microregion remains permanently untreated, I rely on the not-yet-treated parallel trends assumption. Formally, for each dose stratum $d \in \{\text{Low}, \text{High}\}$, treatment cohort g , and post-treatment time $t \geq g$:

$$E[Y_{r,t}(0) - Y_{r,g-1}(0) \mid G_d^r = g] = E[Y_{r,t}(0) - Y_{r,g-1}(0) \mid G_d^r > t]. \quad (5)$$

Three institutional features support the plausibility of this assumption. Treatment timing is driven by the interaction of pre-existing private HEI infrastructure and national scholarship supply rules, not by contemporaneous local wage growth. Comparison units are drawn from the same period but treated later, so they share structural similarities with the treated cohort. And the four-year lag in $D_{r,t}$ (Equation 2) ensures treatment status is predetermined with respect to current labour market conditions. The assumption is assessed empirically through pre-treatment event-study coefficients and a joint Wald test (Section 4.5.1).

Assumption A2 (No Interference / SUTVA). The potential outcome of microregion r depends only on its own treatment status and dose, not on the treatment assigned to other microregions. A violation would arise if ProUni graduates systematically migrate from treated to not-yet-treated regions, contaminating comparison-group outcomes. The use of microregions as the unit of analysis, officially defined by IBGE as self-contained commuting zones, substantially mitigates this risk by internalizing local labor flows. While microregions effectively internalize daily commuting, they do not perfectly internalize permanent collegiate migration. If high-ability students migrate from Low-Dose to High-Dose microregions to utilize scholarships, the resulting spatial sorting will bias the High-Dose ATT downwards (via congestion) and the Low-Dose ATT downwards (via brain drain). Consequently, the microregional estimates should be interpreted as the spatial equilibrium effect of local scholarship allocation (an Intention-to-Treat

parameter), incorporating both the direct human capital shock and the policy-induced geographic reallocation of labor.

Assumption A3 (No Anticipation). Microregions do not adjust their economic behavior in anticipation of future ProUni scholarship allocations. ProUni was publicly announced by the Lula administration in *late 2004* and the first scholarship cohort entered private HEIs in the first semester of 2005. Strictly speaking, this short announcement window means that the earliest years of the panel (2005–2008) could in principle contain limited anticipation effects, especially through HEI capacity planning. Two institutional features make material bias on this channel unlikely. First, the outcome variable reflects *realised* formal wages earned by cohorts aged 22–35, and the four-year lag in the dose variable $D_{r,t}$ (Equation 2) mechanically delays any labor-market consequence until the 2009 panel year at the earliest, by which time the 2004 announcement is firmly in the past. Second, scholarship *volume* at the microregion level depends on two decisions that are not locally forecastable: national budget appropriations in Brasília and private HEI participation choices, neither of which correlate with microregion-level wage trends. As a diagnostic, the pre-treatment event-study coefficients are reported with simultaneous confidence bands (Section 4.5.1) and provide a direct visual test for anticipation: a gradual positive pre-trend would be its natural signature. The joint Wald pre-trend test fails to reject the null at conventional levels for both dose strata, consistent with A3 being at most a minor concern in this design.

Assumption A4 (No Confounding Contemporaneous Policies). No policy other than ProUni should differentially affect treated and comparison microregions in a manner correlated with treatment timing. The two major concurrent policies are FIES (credit expansion) and the Lei de Cotas (public university quota law). Both operate at the national level, meaning their aggregate macroeconomic impacts are absorbed by year fixed effects. However, if FIES loans expanded disproportionately in early-treated ProUni markets, a residual confound could remain. I address this interpretive concern in Section 4.5.9.

A note on dose-response interpretation. A critical distinction established by Callaway et al. (2024) is that the standard parallel trends assumption (A1) is sufficient for identifying the *level effect* $ATT(d | d)$, the average effect of ProUni for microregions that actually received dose d , but is *insufficient* for causal dose-response comparisons across strata. Comparing $ATT(\text{High} | \text{High})$ with $ATT(\text{Low} | \text{Low})$ conflates the causal effect of dose with selection: microregions that receive higher doses may differ systematically in their potential treatment effect functions. Identifying clean dose-response curves requires a stronger

“strong parallel trends” assumption that restricts treatment effect heterogeneity across dose groups.⁷ I therefore treat the dose-specific ATTs as the primary causal results. Cross-dose contrasts are used descriptively and interpreted as suggestive patterns unless explicitly justified under the stronger assumption.

4.4.2 Estimation

For each intensity bin $d \in \{\text{Low}, \text{High}\}$, I estimate group–time average treatment effects, $ATT_d(g, t)$, which compute the average effect for microregions first treated in year g at time t , relative to the not-yet-treated comparison group C_t .

Outcome 1: Log Wages (Intensive Margin). The primary specification, estimated via `did::att_gt()`, targets:

$$ATT_d^w(g, t) = E[\overline{\ln w_{r,t}}(1) - \overline{\ln w_{r,t}}(0) \mid G_d^r = g], \quad (6)$$

where $\overline{\ln w_{r,t}}$ is the population-weighted average of individual log monthly remuneration (in minimum wages) for formal workers aged 22–35 in microregion r and year t (Equation 1). Standard errors are clustered at the microregion level using the multiplier bootstrap (see below).

Outcome 2: Formal Employment Count (Extensive Margin). An analogous specification targets the extensive-margin effect:

$$ATT_d^N(g, t) = E[\ln N_{r,t}(1) - \ln N_{r,t}(0) \mid G_d^r = g], \quad (7)$$

where $\ln N_{r,t}$ is the log number of formal *vínculos* among individuals aged 22–35 in microregion r at time t .

Dynamic Aggregation (Event Studies). To assess the dynamic evolution of the effect and visually test Assumption A1, the $ATT_d(g, t)$ parameters are aggregated into an event-study framework relative to the treatment year g :

$$ATT_{ES}^d(e) = \sum_g w_g ATT_d(g, g + e), \quad (8)$$

where $e = t - g$ is event time and w_g are cohort-size weights. This is implemented via `did::aggte(type = "dynamic")`.

⁷Formally, strong parallel trends requires that the untreated potential outcome paths are parallel not just within dose groups but *across* them. See Assumption 3 in Callaway et al. (2024).

Outcome 3: Cohort Wage Gap (Triple-Difference). The primary outcome of the age-cohort analysis is the within-microregion wage gap between the young and old cohorts:

$$\Delta w_{r,t} = \bar{w}_{r,t}^{\text{young}} - \bar{w}_{r,t}^{\text{old}}, \quad (9)$$

where $\bar{w}_{r,t}^c$ is the average log wage for cohort c in microregion r at time t . Any local shock that affects all age groups equally, such as a factory closure, commodity price change, or macroeconomic fluctuation, cancels in the differencing, leaving only the age-specific ProUni channel. This yields an estimating equation analogous to Equation 6:

$$\Delta w_{r,t} = ATT_d^\Delta(g, t) \cdot \mathbf{1}\{G_d^r = g, T = t\} + \alpha_r + \lambda_t + \varepsilon_{r,t}, \quad (10)$$

where a positive ATT_d^Δ indicates that ProUni raised the relative wage of the exposed (young) cohort. Additionally, estimating Equations 6 separately for each age group provides a decomposition. If the young-only ATT is positive and the old-only ATT is zero, the effect is consistent with an age-specific channel. If both are positive, the pattern is consistent with positive general-equilibrium spillovers, although residual confounding cannot be excluded mechanically.

Outcome 4: First Stage, College Attainment Gap. To recover the implied return to a ProUni degree, I estimate the effect of scholarship density on the college attainment gap:

$$\Delta s_{r,t} = ATT_d^s(g, t) \cdot \mathbf{1}\{G_d^r = g, T = t\} + \alpha_r + \lambda_t + \varepsilon_{r,t}, \quad (11)$$

where $\Delta s_{r,t} = s_{r,t}^{\text{young}} - s_{r,t}^{\text{old}}$ is the difference in college-educated worker shares between cohorts. The college-educated share is computed from the RAIS education variable (`grau_instrucao_apos_2005` ≥ 9 , capturing workers with completed tertiary education or above). The first-stage effect of programme intensity on the formal-sector college share, estimated from equation (11), is small in magnitude and statistically imprecise. The implied Wald/LATE estimate is therefore not reported as a primary causal object. The remainder of the analysis is interpreted as reduced-form evidence on the local labour-market response to ProUni exposure, leaving the individual return to a scholarship for future work that links beneficiary identifiers to administrative wage histories.

Heterogeneity Subsamples. To investigate distributional impacts, Equations 6 and 7 are re-estimated on mutually exclusive demographic subsamples defined by Gender (Female vs. Male), Ethnicity (Non-white vs. White), and Region

(North/Northeast vs. South/Southeast/Centre-West). For each subgroup, the outcome variable is computed strictly from the corresponding subset of RAIS records, maintaining the same cohort definitions and comparison groups.

Covariate selection. The doubly-robust specifications condition on pre-treatment (2010 Census) youth population, log household income per capita, literacy rate, and share of the 18–24 population enrolled in school, together with a 2009 snapshot of private-HEI density and HE enrolment intensity from the INEP Education Census. Literacy rate, schooling share, and HE density are outcomes of the same regional development process that determined ProUni uptake, and conditioning on such variables can in principle reintroduce omitted-variable bias (Cinelli, Forney, & Pearl, 2022). Three features of the design mitigate this risk. Every covariate is measured before the first year in which the four-year-lagged dose variable can be non-zero (2009), so none can mechanically mediate the treatment effect. The doubly-robust estimator of Sant’Anna and Zhao (2020), implemented in `did::att_gt()` via the explicit argument `est_method = "dr"`, achieves consistency as long as *either* the outcome regression *or* the propensity-score model is correctly specified, providing protection against individual covariate misspecification. And the covariate set was determined ex ante from the selection mechanism documented by McCowan (2007): private HEIs concentrate in wealthier, more educated markets, so controlling for wealth and educational infrastructure targets the precise selection channel of concern. Specifications that drop potentially endogenous covariates are reported in the appendix.

The τ_{20} treatment-timing threshold. A microregion’s treatment year is defined as the first year in which the four-year-lagged dose $D_{r,t}$ exceeds the 20th percentile τ_{20} of the 2019 cross-sectional dose distribution (Equation 3). Using a zero cutoff would conflate microregions that received a single pilot scholarship in 2005 with those that received hundreds by 2010, introducing measurement error in treatment timing that would bias the Callaway–Sant’Anna ATTs toward zero. Using τ_{20} instead requires meaningful programme penetration before a microregion enters the treated cohort and matches the policy-relevant notion of “substantive exposure” rather than nominal contact. Sensitivity analyses in the appendix re-estimate the main specifications under alternative thresholds τ_{25} and τ_{33} ; the low-dose scarcity pattern is qualitatively preserved across all three cutoffs (Table 9).

Falsifiability. The hypothesis that ProUni raised formal-sector wages in moderately exposed microregions is falsifiable in four ways, each mapped to a test

reported in Section 5: monotonically upward Low-Dose pre-treatment event-study coefficients (Assumption A1); a non-zero placebo effect on workers aged 36–50, too old to have been ProUni-eligible; a Low-Dose DR robust confidence set that includes zero at $\bar{M} = 0$ in the HonestDiD sensitivity analysis; and a TWFE/Callaway and Sant’Anna reconciliation in which the two estimators yield statistically indistinguishable coefficients on the same sample. None of these patterns is observed in the data.

4.4.3 Inference

All confidence intervals in the main event-study figures are constructed using the multiplier bootstrap implemented in the `did` package with `bstrap = TRUE` and `cband = TRUE` (the package defaults). This procedure produces *simultaneous* confidence bands that cover the entire path of event-study coefficients with probability $(1 - \alpha)$, accounting for the multiple testing problem inherent in evaluating many group-time ATTs simultaneously.⁸ Standard errors are clustered at the microregion level, which is the level of treatment assignment.

Effective-cluster count. MacKinnon, Nielsen, and Webb (2023) show that cluster-robust inference can suffer severe size distortions when the number of treated clusters is small relative to the total, even when the overall cluster count is large. In the main specification, the Low-Dose stratum contains 167 treated clusters against 112 not-yet-treated comparisons, and the High-Dose stratum contains 140 treated clusters against 112 comparisons. Both treatment arms are therefore well above the ~ 20 effective-cluster threshold below which wild-bootstrap size distortions become severe, and the multiplier bootstrap implemented via `did` is additionally robust to moderate cluster imbalance. A wild-cluster-bootstrap robustness check with 9,999 replications is reported in the appendix as a diagnostic cross-check; the main conclusions are unchanged.

4.4.4 Mechanisms: Testing for General Equilibrium Effects

A key theoretical concern in large-scale education programs is the potential for general equilibrium effects to dampen private returns. As noted by Duflo (2001), a massive supply shock could depress the wage premium through oversupply. This thesis therefore compares the dose-specific event studies as evidence on whether

⁸The multiplier bootstrap uses weights drawn from the influence function representation of the estimator and produces critical values larger than 1.96 (typically ~ 2.7), reflecting the simultaneous coverage requirement. Pointwise confidence intervals (setting `cband = FALSE`) would have 95% marginal coverage but much lower joint coverage and are therefore not reported as the primary inference tool (Callaway & Sant’Anna, 2021).

estimated effects are larger in relatively scarce or highly exposed local markets. Two competing empirical signatures can be articulated ex ante. A pattern consistent with scarcity would return $ATT_{ES}^{\text{Low}}(e) > ATT_{ES}^{\text{High}}(e)$ for post-treatment event times, indicating that wage premia are better preserved when the supply shock is relatively scarce. A pattern consistent with attenuation in highly exposed markets would return an $ATT_{ES}^{\text{High}}(e)$ that is materially smaller or statistically indistinguishable from zero, indicating attenuation through congestion, credential inflation, or compositional dilution.

A critical identification challenge in repeated cross-sectional administrative data is distinguishing true General Equilibrium (GE) wage depression from compositional effects. If ProUni successfully integrates marginalized workers from the informal sector into RAIS, these marginal entrants likely possess lower unobserved labor market experience than inframarginal formal workers. Therefore, a flat or negative $ATT_{ES}^{\text{High}}(e)$ could reflect the mechanical dilution of the average formal wage rather than true credential inflation. The age-cohort decomposition (Outcome 3) partially addresses this by restricting the comparison to similarly aged entrants, but residual compositional bias within the formal sector remains a caveat to the high-dose interpretation.

As discussed above, these cross-dose comparisons are interpretable as causal dose-response effects only under the strong parallel trends assumption. Under standard parallel trends, they reflect a combination of causal dose effects and selection into dose levels.

4.5 Robustness Checks and Validity Tests

The following validation protocol is executed in Section 5 to systematically test the identification assumptions and probe the sensitivity of the main findings to specification choices.

4.5.1 Testing Assumption A1: Pre-Trend Analysis

The primary diagnostic for the parallel trends assumption is the inspection of pre-treatment event-study coefficients $ATT_{ES}^d(e)$ for $e < 0$. Under A1, these should be statistically indistinguishable from zero. I report both point estimates with simultaneous confidence bands and a joint Wald test for pre-trends (the null hypothesis being that all pre-treatment dynamic effects are jointly zero).

4.5.2 Age-Cohort Placebo

As a built-in falsification test, the main specification is estimated separately for the young (22–28) and old (29–35) cohorts. Under the identifying assumption

that ProUni’s human capital channel operates exclusively through the exposed cohort, the old-cohort ATT should be zero. A non-zero old-cohort effect would indicate either (a) a violation of the identifying assumption (local economic shocks correlated with treatment timing) or (b) positive general equilibrium spillovers from the inflow of college-educated workers. The latter interpretation is distinguished from the former by examining whether the old-cohort effect is present only in treated microregions and only after treatment onset.

This within-market placebo is more powerful than an external age-cohort test, for example workers aged 36–50, because the old cohort inhabits the *same microregion-year cell* as the young cohort, perfectly controlling for local conditions. Any remaining differential between young and old must therefore arise from age-specific mechanisms rather than shared local trends.

4.5.3 Testing for Omitted Variable Bias: Continuous TWFE and Doubly Robust Covariates

Given the explicit geographic concentration of ProUni in Brazil’s wealthier South and Southeast, there is a risk that simple group-time estimators confound the causal effect of ProUni with regional macroeconomic shocks, for example the 2014–2016 Brazilian recession. To test whether treatment is confounded by pre-existing regional wealth, I employ two strategies. First, I estimate a sequence of continuous treatment TWFE models (Linear, Logarithmic, and Quadratic doses) interacting the continuous $D_{r,t}$ with a post-treatment dummy to detect non-linear attenuation patterns.

Second, to address omitted variable bias, I estimate a covariate-adjusted TWFE model and a Doubly-Robust Callaway & Sant’Anna specification. These models interact the post-treatment dummy with baseline (2010 Census) microregion characteristics, notably the local youth population, the share of non-white workers, and the average household income per capita. If the inclusion of these covariates, particularly baseline wealth, absorbs the negative effects observed in High-Dose microregions, this would suggest that the raw high-dose penalty partly reflects macroeconomic contractions disproportionately hitting wealthy geographic hubs rather than a pure educational saturation effect.

4.5.4 Placebo Cohort Test

As a falsification exercise, I re-estimate the main wage models on the sample of formal workers aged 36–50 years, a cohort too old to have been directly exposed to ProUni at standard university-going age. If the main results capture a genuine human capital supply shock from ProUni graduates entering the labor market,

this older cohort should show no significant post-treatment effects. Conversely, non-zero effects in the placebo cohort would indicate that the baseline results reflect broad local economic shocks correlated with treatment timing rather than the specific mechanism of skill supply expansion.

4.5.5 Discretization Sensitivity

To assess whether results depend on the chosen dose thresholds, I re-run the main wage models under two alternative partition schemes: (i) τ_{25}/τ_{75} , which narrows the Low bin and sharpens the treatment–comparison contrast; and (ii) τ_{33}/τ_{67} , which provides a symmetric tercile split with more balanced bin sizes. Table 9 in the appendix reports the point estimates across all three cutoff schemes. The sign of the Low-Dose wage effect is preserved across cutoffs, but the alternative-cutoff specifications do not pass the pre-trend test cleanly: the joint Wald pre-trend p -value is 0.097 for Low/q25_75 and 0.093 for Low/q33_67, both well below the $p = 0.327$ obtained under the baseline $\tau_{20}/\tau_{50}/\tau_{75}$ partition. The Low/Emp (employment) specification likewise returns a pre-trend p -value of 0.115, weaker than the wage pre-trend. These patterns should be read as partial caveats to the identification story rather than as falsifications: the sign and sign-significance of the Low-Dose wage effect are preserved under the alternative cutoffs, but the parallel-trends assumption is more firmly supported by the baseline partition than by the alternatives, and this should temper readings of the alternative-cutoff point estimates as unqualified robustness evidence.

4.5.6 Sensitivity to Parallel-Trends Violations: HonestDiD

Beyond testing whether pre-trends are statistically distinguishable from zero, I report the partial-identification sensitivity analysis of Rambachan and Roth (2023). Given an event-study estimate $\widehat{ATT}_{ES}^d(e)$, the **HonestDiD** procedure constructs the set of post-treatment effects that remain consistent with the data once one allows the underlying parallel-trends violation to be at most a scalar multiple $\bar{M}$ of the largest observed pre-treatment violation. Rather than a single point estimate, the diagnostic is the *breakdown* $\bar{M}$: the smallest value at which zero enters the robust 95% confidence set. A small breakdown $\bar{M}$ indicates fragility to unobserved pre-trends; a larger value indicates that the result survives even generous deviations from parallel trends. I report breakdown values for the Low-Dose wage ATT and the Low-Dose DR specification in the appendix. High-Dose HonestDiD analysis is omitted because its near-zero point estimate makes breakdown $\bar{M}$ uninformative.

4.5.7 Alternative Lag Specification

The baseline specification assumes a four-year lag between scholarship allocation and labor market entry. As a sensitivity check, I re-estimate all models with a three-year lag, which assumes faster degree completion. The lag sensitivity test serves a dual purpose: it probes the plausibility of the timing assumption and provides indirect evidence on whether results are sensitive to changes in the effective treatment window.

4.5.8 TWFE Benchmark

To illustrate the empirical relevance of heterogeneous treatment effect bias, I estimate the naive TWFE specification:

$$\overline{\ln w_{r,t}} = \beta_{\text{TWFE}} \cdot \mathbf{1}\{D_{r,t} > \tau_{20}\} + \alpha_r + \lambda_t + \varepsilon_{r,t}, \quad (12)$$

with standard errors clustered at the microregion level, implemented via `fixest::feols()`. As shown by Goodman-Bacon (2021), $\hat{\beta}_{\text{TWFE}}$ is a weighted average of all 2×2 comparisons, including “forbidden” comparisons where already-treated units serve as controls. If treatment effects are heterogeneous across cohorts or evolve over time, some of these weights can be negative, attenuating or sign-reversing the TWFE estimate relative to the properly aggregated Callaway–Sant’Anna ATTs. A substantial divergence between the two estimates validates the methodological choice of the heterogeneity-robust estimator.

4.5.9 Concurrent Policy Controls

FIES and the Federal Quota Law (Lei 12.711/2012) constitute important concurrent shocks in the higher education market, together with the REUNI federal university expansion (Decreto 6.096/2007). Since all three policies are national in scope and are not designed to target microregions based on their ProUni scholarship density, their common impact is absorbed by year fixed effects and does not mechanically confound the within-microregion event-time variation. To move beyond this aggregate argument, I construct three microregion-level covariates that measure differential exposure to each confounder: (i) pre-treatment FIES loan density, built from MEC’s FIES microdata aggregated to the 2005–2009 window and divided by the 2010 Census youth population; (ii) a REUNI exposure measure, $\log(1 + \text{n_federal_heis}_{2009})$, capturing the stock of federal higher-education institutions before the federal expansion peaked; and (iii) the 2012 federal share of scholarships, a proxy for the intensity of Lei 12.711 quota eligibility. I then re-estimate the Low-Dose and High-Dose DR specifications including these three

covariates in `xformula`, and separately report a TWFE specification in which time-varying FIES loan density enters additively. Results are reported in the appendix and leave the core Low-Dose scarcity pattern intact.

4.5.10 HEI Linkability and Quality Robustness

A residual concern is that ProUni scholarship volume is mechanically correlated with the *quality* of the local private HEI stock: microregions whose scholarships come predominantly from low-quality institutions could register a nominal “dose” without generating the human-capital inflow the design presumes. To probe this, I link the ProUni microdata to the INEP Higher Education Census via the `id_ies` institutional identifier and merge in each HEI’s Índice Geral de Cursos (IGC) and Conceito Preliminar de Curso (CPC) quality scores, snapshotted in the pre-treatment 2009 window. I then construct an HEI-level first-stage diagnostic (scholarships per enrolment per HEI-year) and, as a robustness check, re-estimate the Low-Dose wage ATT dropping microregions in which at least 50% of ProUni scholarships originate from HEIs in the bottom quartile of the CPC distribution. The resulting point estimate is reported alongside the baseline in Table ?? and discussed in Section 5.

4.5.11 Bacon Decomposition

The Goodman-Bacon (2021) decomposition partitions the TWFE estimand into weighted two-by-two comparisons and classifies each as a “good” comparison (not-yet-treated serving as control) or a “forbidden” comparison (already-treated serving as control for a later-treated unit). I report the aggregate weights attached to each class in Appendix Table 15. The forbidden-comparison weight is non-trivial in the ProUni panel, which is consistent with the documented divergence between the TWFE estimator and the Callaway–Sant’Anna aggregation and provides an independent indicator that heterogeneous treatment effects are present. The decomposition is reported as a diagnostic on the naive TWFE specification, not as a primary causal estimator.

4.5.12 Five-Region Belíndia Heterogeneity

The baseline two-macro-region split (North/Northeast vs. South/Southeast/Centre-West) sacrifices information on the Centre-West’s distinct agribusiness growth trajectory. I therefore re-estimate the main wage ATT separately for each of the five IBGE macro-regions (North, Northeast, Centre-West, Southeast, South), reported in Table 16. The resulting within-region estimates provide a direct test

of whether the scarcity premium identified in Low-Dose microregions is a genuine across-the-board phenomenon or is instead driven by a single macro-region’s growth dynamics.

5 Results

This chapter presents the empirical findings in five stages. Section 5.1 establishes the geographic structure of treatment variation. Section 5.2 reports the primary doubly-robust (DR) event-study estimates for wages as the headline causal results. Section 5.3 elevates the TWFE vs. Callaway-Sant’Anna comparison as a core methodological finding. Section 5.4 decomposes effects using the Duflo-style age-cohort design. Section 5.5 explores distributional impacts by gender and ethnicity, and Section 5.6 consolidates all robustness checks.

Before turning to the estimates, it is useful to distinguish clearly between what is identified and what is more tentative. The primary causal objects in this chapter are the within-stratum group-time ATTs for Low-Dose and High-Dose microregions estimated under the not-yet-treated parallel trends assumption, with the doubly-robust specification serving as the headline. By contrast, direct comparisons between the Low-Dose and High-Dose paths require the stronger assumption of Callaway et al. (2024) and are therefore interpreted as suggestive patterns rather than as clean causal dose-response effects.

5.1 Geographic Distribution of Treatment Intensity

Figure 4 displays the spatial distribution of cumulative ProUni scholarship density ($D_{r,2019}$) across Brazil’s 558 microregions. The map reveals a pronounced geographic concentration: the South and Southeast, particularly the São Paulo metropolitan corridor, Porto Alegre, and Belo Horizonte, exhibit the highest treatment intensities, while the Amazon basin and parts of the Northeast received substantially lower exposure. This pattern is consistent with the unequal distribution of pre-existing private HEI infrastructure, although the map by itself does not establish exogeneity.

Table 1 reports summary statistics by dose bin. High-Dose microregions are characterized by larger youth populations and higher absolute treatment intensities, having crossed the treatment threshold earlier (median year 2012) than their Mid-Dose (median 2013) and Low-Dose counterparts (median 2017). The Mid-Dose stratum occupies an intermediate position on both dimensions: 139 microregions with average youth populations and absolute dose densities falling between the two extremes. These differences reinforce the need to interpret cross-

ProUni Scholarship Density by Microregion

Cumulative scholarships per 1,000 youth (lagged 4 years), 2019

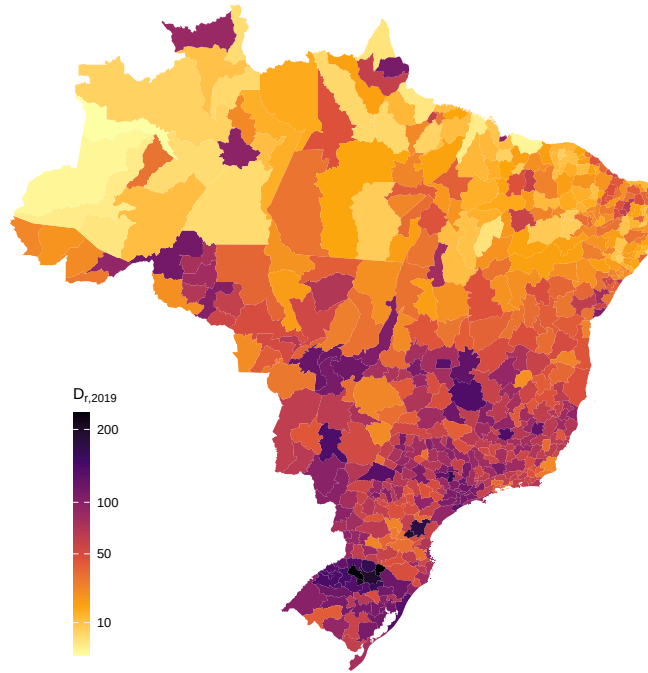


Figure 4: **ProUni Scholarship Density by Microregion, 2019.** Cumulative scholarships per 1,000 youth (aged 18–24), lagged four years. Color intensity follows a square-root scale to accommodate the right-skewed distribution.

dose contrasts cautiously, since heavily exposed markets are not simply scaled-up versions of lightly exposed ones.

Table 1: **Summary Statistics by Dose Bin (2019)**

Dose Bin	N. Regions	Avg. Youth Pop.	Avg. Density	Median Treatment Yr
Low	167	27,689	33.74	2017
Mid	139	48,120	61.02	2013
High	140	74,598	107.43	2012

5.2 Primary Results: Doubly-Robust Estimates

5.2.1 Wages: Headline Doubly-Robust Evidence

The central empirical result of this thesis is a *reduced-form* doubly-robust estimate for Low-Dose microregions: the average log formal wage of workers aged 22–35 is +3.04% ($SE = 0.0183$, $p = 0.097$) higher in treated microregions than in the not-yet-treated comparison group, averaged over the first four post-treatment

years. The estimate is positive and marginally distinguishable from zero at the ten-per-cent level, but its magnitude is modest and its precision does not rule out effects in the 0–7 per cent band. The estimand is the *average treatment effect on the treated* (ATT) in the moderate-exposure stratum, not the average treatment effect that would be observed if ProUni were reallocated to currently-untreated microregions; extrapolating the point estimate to the latter policy-relevant quantity would require the stronger conditional-exogeneity assumption that treatment dose is independent of untreated potential outcomes given observables. The ATT interpretation is nonetheless directly policy-relevant for the decision of whether to *maintain* existing allocations in moderate-exposure areas. The DR specification conditions on pre-treatment (2010 Census) youth population, log household income per capita, literacy rate, schooling share, and a pre-treatment (2009) snapshot of HE density and enrolment intensity from the INEP Education Census, via the `xformula` argument of `att_gt()`, following the doubly-robust identification strategy of Sant’Anna and Zhao (2020). The unconditional simple ATT (+1.28%, $SE = 0.0116$) is reported as a robustness check; the fact that the DR estimate is larger than the unconditional ATT indicates that baseline regional income is a meaningful confounder and that the bias in a naive specification runs toward zero.

Figure 5 presents the baseline wage event studies. The dynamic estimates suggest a divergence between Low-Dose and High-Dose microregions after treatment onset, but the safest interpretation remains within each stratum rather than between strata.

Figure 5 reveals a clean pre-treatment pattern for both dose bins: event-study coefficients for $e < 0$ fluctuate around zero with no systematic drift, and the joint Wald test fails to reject the null of zero pre-trends ($p = 0.327$ for Low Dose; $p = 0.881$ for High Dose). Non-rejection supports the identifying assumption, though it does not constitute proof of it.

Post-treatment, the Low-Dose series displays a gradually increasing trajectory reaching roughly 2.5–3.5 per cent by event time $e = 5$ –6. The DR estimate of 3.04% indicates that this pattern is not an artefact of compositional differences between treated and comparison regions; the event window is trimmed to $e \leq 4$ for the DR specification to maintain adequate propensity-score overlap, and the simultaneous confidence bands use critical values of approximately 2.7 to maintain correct joint coverage. The High-Dose series, by contrast, remains close to zero throughout the post-treatment window (overall ATT = -0.70% , $SE = 0.0069$). As shown in Section 5.6, this penalty weakens once macroeconomic geography is controlled for, so it should not be read as definitive evidence of saturation.

The Mid-Dose stratum ($N = 139$ microregions, median treatment year 2013)

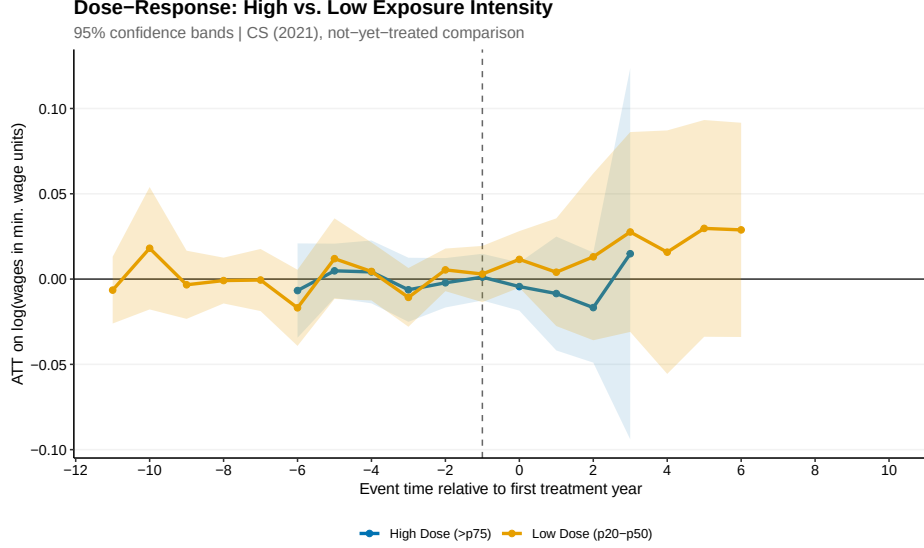


Figure 5: **Dose-Response: High vs. Low Exposure Intensity.** Dynamic ATT estimates from Callaway and Sant’Anna (2021), aggregated via `aggte(type = “dynamic”)`. Shaded regions denote 95% simultaneous confidence bands from the multiplier bootstrap. The dashed vertical line marks $e = -1$, the last pre-treatment period.

delivers a statistically significant negative estimate: overall $ATT = -1.57\%$ ($SE = 0.0088$, $p = 0.074$). Read jointly with the positive Low-Dose estimate and the near-zero High-Dose estimate, the three-stratum pattern is therefore not monotonic: the reduced-form wage effect moves from positive in moderately exposed markets, through negative in intermediately exposed markets, to statistically indistinguishable from zero in the most heavily exposed ones. This non-monotonicity rules out a simple “diminishing returns” reading of the dose-response curve and instead points to a more intricate interaction between dose intensity, local skilled-labour absorption, and the macroeconomic trajectory of the 2014–2016 recession, which disproportionately loaded onto the Mid-Dose and High-Dose microregions that peaked in ProUni exposure during the downturn.

5.2.2 Employment: Extensive Margin

Figure 7 reports the employment-margin estimates. Both dose bins show near-zero effects for the first four post-treatment years. The overall ATT on log formal employment is 1.82% ($SE = 0.0332$) for Low Dose and 0.65% ($SE = 0.0396$) for High Dose.

The absence of large employment effects is not surprising in a context where RAIS captures only formal employment. ProUni may shift workers from informal to formal jobs without changing the total headcount of 22–35 year-olds, so the employment outcome should be interpreted as a formal-sector extensive-margin

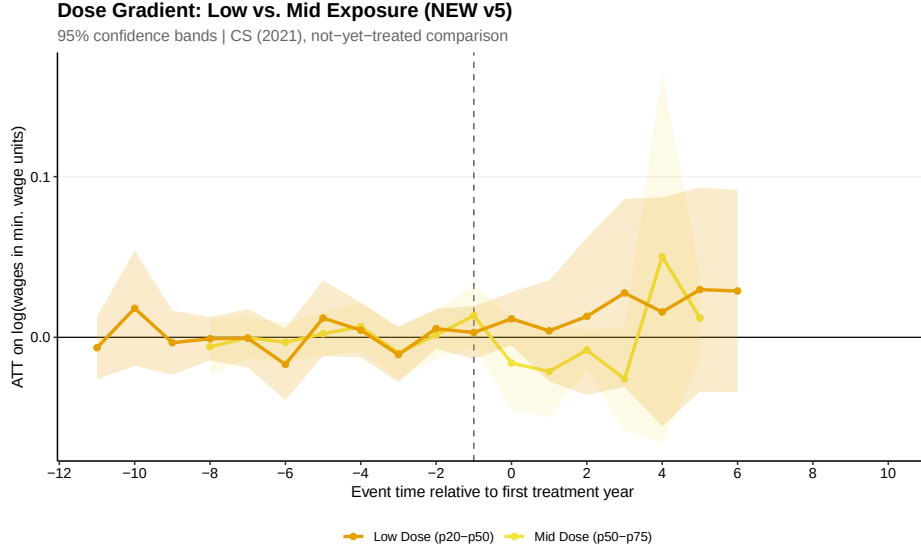


Figure 6: **Mid-Dose Event Study.** Callaway–Sant’Anna dynamic ATT for the Mid-Dose stratum ($\tau_{50}-\tau_{75}$, $N = 139$ microregions). The negative post-treatment trajectory contrasts with the positive Low-Dose pattern and near-zero High-Dose pattern, establishing the non-monotone dose–response.

measure rather than as a complete labour-market quantity effect.

5.3 TWFE Bias Demonstration

A central methodological contribution of this thesis is demonstrating the empirical relevance of heterogeneous-treatment-effect bias in this particular setting. Table 2 compares the naïve binary TWFE estimates with the Callaway and Sant’Anna (2021) ATTs. The TWFE wage coefficient is -0.65% ($SE = 0.0037$, $p = 0.079$), negative and marginally significant, in the opposite direction to the Low-Dose CS estimate. The TWFE employment coefficient likewise masks the positive low-dose extensive-margin pattern. This divergence is not a sampling artefact; it is precisely the forbidden-comparison bias documented theoretically by Goodman-Bacon (2021) and de Chaisemartin and D’Haultfœuille (2020): early-treated, High-Dose (and therefore larger, wealthier) microregions enter as implicit controls for later-treated units, contaminating the pooled TWFE coefficient with negative comparisons. The CS estimator avoids this by restricting comparisons to not-yet-treated units within each dose stratum.

The $+3.69$ percentage-point gap between the TWFE and the DR Low-Dose estimate (-0.65% vs. $+3.04\%$) provides a concrete quantification of the bias magnitude in this application. This result reinforces the recommendation of Callaway and Sant’Anna (2021) to use heterogeneity-robust estimators in staggered designs, and constitutes an independent empirical contribution of this thesis beyond the ProUni-specific findings.

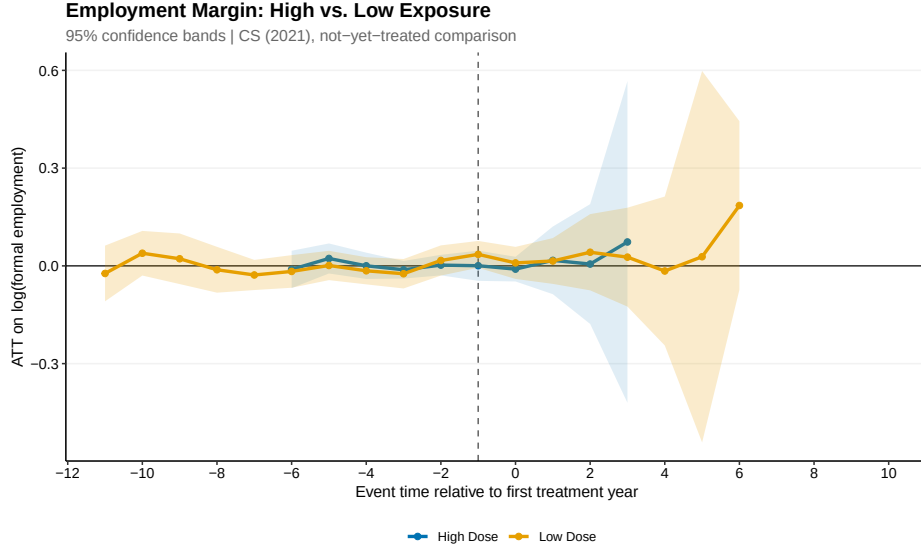


Figure 7: **Employment Margin: High vs. Low Exposure.** ATT on log formal employment (vínculos, ages 22–35).

Table 2: **TWFE vs. Callaway-Sant’Anna: Empirical Illustration of Forbidden-Comparison Bias**

Specification	ATT	SE	N
TWFE / Wages (pooled)	−0.0065*	0.0037	7,812
CS / Low Dose (wages, unconditional)	0.0128	0.0116	3,906
CS / Low Dose (wages, DR)	0.0304*	0.0183	3,906
CS / High Dose (wages)	−0.0070	0.0069	3,528
TWFE / Wage Gap (cohort)	0.0072	0.0042	7,810
CS / Gap Low	−0.0054	0.0120	3,904
CS / Gap High	0.0234	0.0273	3,526

* $p < 0.10$. Standard errors clustered by microregion.

5.4 Age-Cohort Decomposition

5.4.1 Headline: Cohort Wage Gap

To isolate the age-specific human capital channel, I estimate the treatment effect on the within-microregion wage gap between the exposed cohort (ages 22–28) and the control cohort (ages 29–35), as described in Section 4.4.2. Figure 8 reports the results.

The cohort wage-gap results are noisier than the pooled estimates and do not deliver a sharp confirmation of the baseline low-dose pattern. The High-Dose series shows a modest positive drift after treatment (overall ATT of 2.34%, $SE = 0.0273$), while the Low-Dose series remains close to zero and slightly negative on average (overall ATT of −0.54%, $SE = 0.0120$). Because the confidence intervals are wide, these estimates are best viewed as diagnostic evidence on mechanisms rather than as stand-alone headline effects.

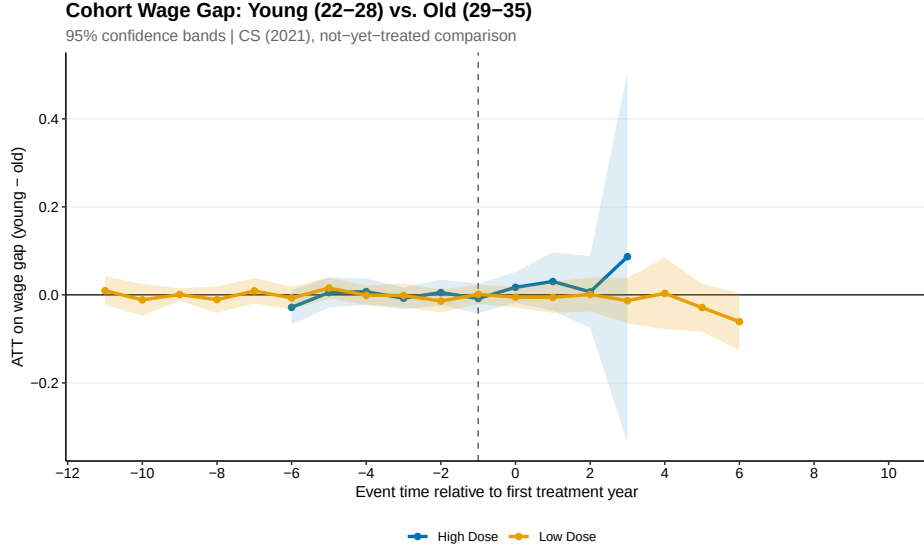


Figure 8: **Cohort Wage Gap: Young (22–28) vs. Old (29–35).** ATT on the within-microregion wage gap ($\Delta w_{r,t}$). A positive ATT indicates that ProUni raised the relative wage of the young (exposed) cohort.

5.4.2 Decomposition: Young-Only vs. Old-Only

To understand what drives the cohort wage-gap dynamics, I estimate the ATT separately for each age group. Figures 9 and 10 report these decompositions.

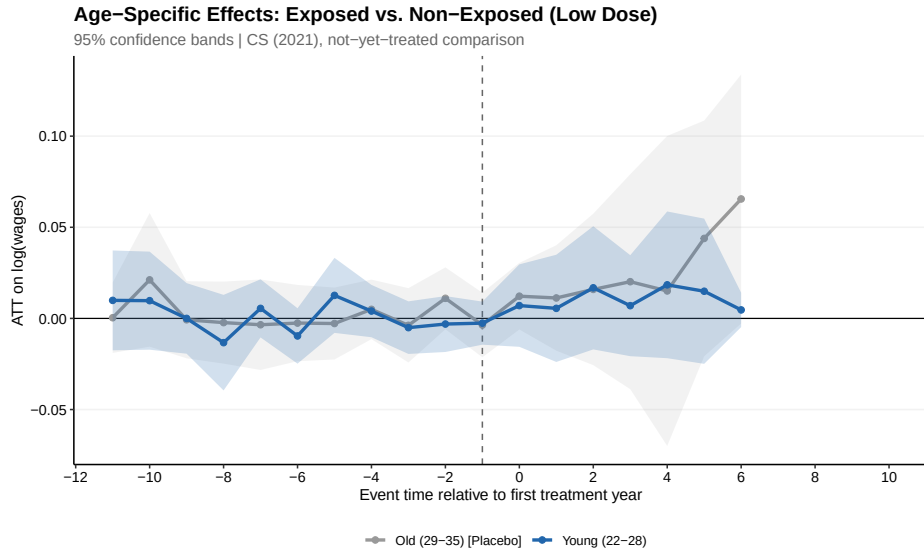


Figure 9: **Age-Specific Effects: Low Dose.** Separate ATTs for young (22–28, exposed) and old (29–35, control cohort).

The decomposition reveals suggestive but not definitive patterns. In Low-Dose microregions (Figure 9), the young cohort shows a modest positive trajectory, but the old cohort also rises. One interpretation is the presence of positive general-equilibrium spillovers: a moderate inflow of college-educated workers may raise

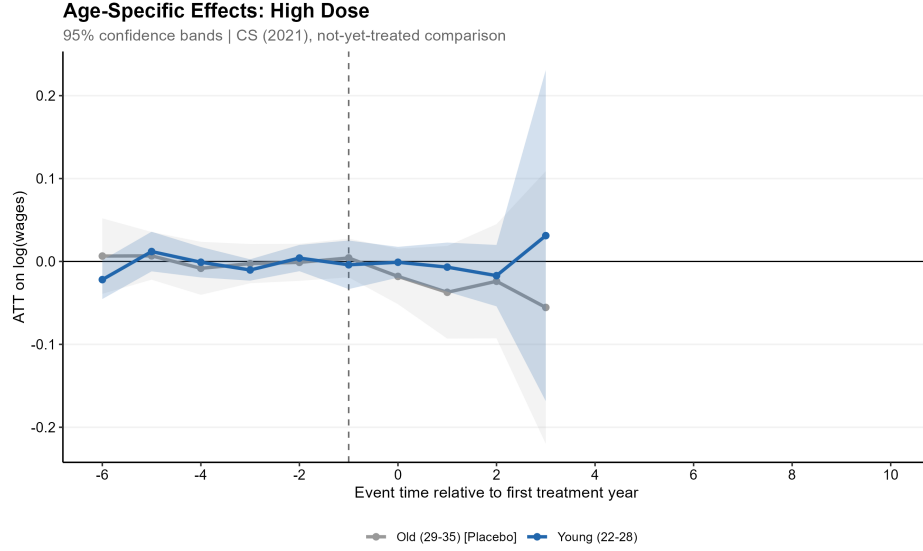


Figure 10: **Age-Specific Effects: High Dose.** Separate ATTs for young (22–28) and old (29–35).

local productivity in a way that benefits multiple age groups. A more cautious alternative is that some residual local shocks correlated with treatment timing remain in the data. The evidence is therefore consistent with spillovers, but it does not mechanically prove them.

In High-Dose microregions (Figure 10), the young cohort remains close to zero while the old cohort declines. This pattern is consistent with displacement or relative wage compression for older workers in heavily exposed markets. At the same time, because the old cohort is intended as a placebo, non-zero effects on that group also warn against an overly mechanical causal interpretation.

5.4.3 First Stage: College Attainment Gap

Figure 11 reports the estimated effect of ProUni density on the college share gap between young and old workers.

The first-stage estimates are exceptionally small in magnitude for most post-treatment periods. For High-Dose microregions, the overall ATT is 0.24% ($SE = 0.0031$), while for Low-Dose it is effectively zero. The weakness of the first stage likely reflects both measurement error in employer-reported RAIS education and dilution from computing tertiary shares over the full formal workforce. The reduced-form analysis therefore proceeds without recourse to a Wald/LATE rescaling.

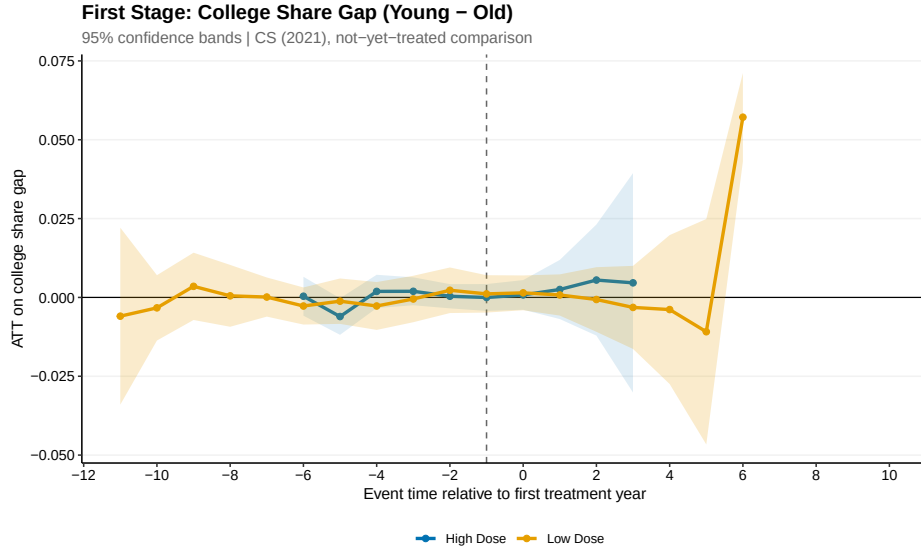


Figure 11: **First Stage: College Share Gap (Young – Old)**. ATT on the difference in college-educated worker shares between age cohorts.

5.5 Heterogeneity by Gender and Ethnicity

5.5.1 Pooled Heterogeneity

Gender. Figure 12 reports the gender-stratified pooled event studies. In Low-Dose microregions, male workers exhibit a positive and growing wage effect (overall ATT 1.67%, $SE = 0.0169$), while female workers remain close to zero on average (overall ATT -0.03% , $SE = 0.0008$). In High-Dose microregions, both genders show estimates near zero.

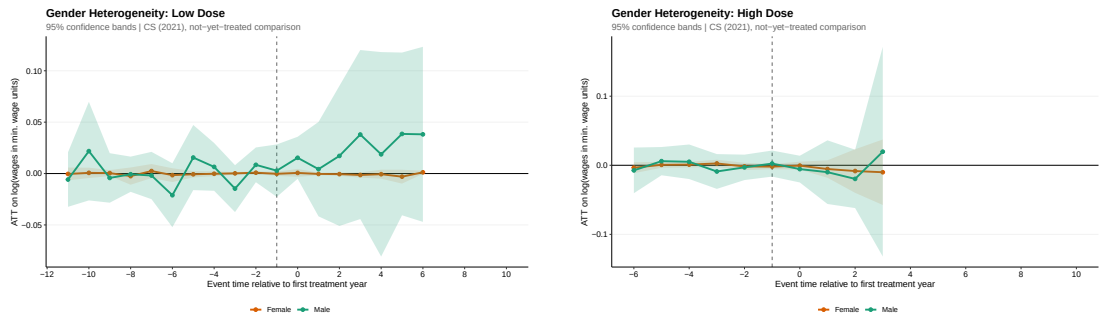


Figure 12: **Gender Heterogeneity in Pooled Wage Effects**. Left: Low Dose. Right: High Dose.

These estimates are consistent with a male wage advantage in the conversion of educational expansion into formal-sector earnings. However, the evidence should be interpreted as reduced-form heterogeneity rather than as direct proof of a single mechanism such as field sorting, discrimination, or labour-force attachment.

Ethnicity. Figure 13 shows that in Low-Dose microregions, non-white workers exhibit positive trajectories (overall ATT 1.28%, $SE = 0.0123$), while white workers remain close to zero on average (overall ATT -0.22% , $SE = 0.0095$). In High-Dose microregions, both groups again show near-zero effects.

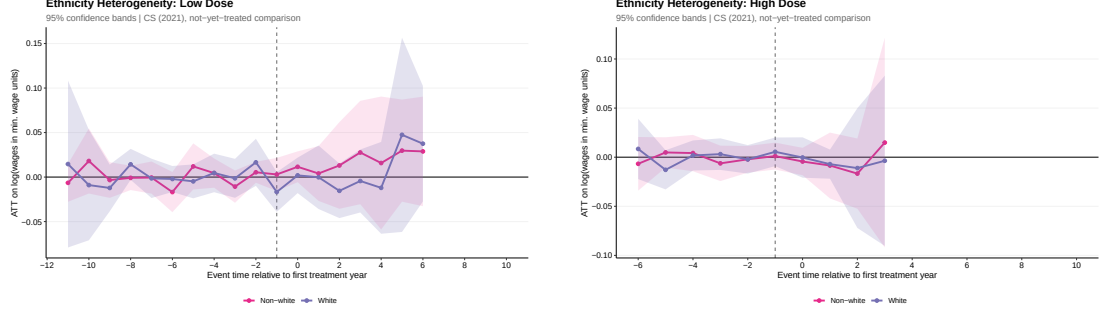


Figure 13: **Ethnicity Heterogeneity in Pooled Wage Effects.** Left: Low Dose. Right: High Dose.

These patterns are consistent with ProUni improving formal-sector wage outcomes for non-white workers in moderately exposed markets, coherent with the programme’s affirmative-action design. Even so, the thesis cannot claim full distributive parity on the basis of these reduced-form estimates alone.

5.6 Robustness Checks

5.6.1 Continuous TWFE and Omitted Variable Bias

To test whether the High-Dose penalty is confounded by macroeconomic geography, I estimate a sequence of continuous-treatment TWFE models interacting the continuous dosage measure $D_{r,t}$ with a post-treatment dummy.

Table 3 summarizes the results. An unconditional linear specification (Model 1) yields a marginally significant negative dose-response coefficient ($\beta = -0.00013$, $p = 0.0539$). A logarithmic transformation (Model 2) provides a somewhat better fit ($\beta = -0.0021$, $p = 0.0480$). A quadratic specification (Model 3) shows a significant negative linear term but an insignificant squared term ($p = 0.3131$). Once the TWFE model is augmented with baseline covariates interacted with the post-treatment dummy (Model 4), the direct effect of ProUni dose disappears ($p = 0.3699$), while the interaction between post-treatment timing and baseline household income per capita is large, negative, and statistically significant ($\beta = -0.0377$, $p = 0.0015$). This confirms that unconditional high-dose penalties partly reflect adverse macroeconomic shocks concentrated in Brazil’s wealthier hubs during the 2014–2016 recession, not a pure educational saturation effect.

Table 3: **Continuous TWFE Models: Testing for Saturation and OVB**

	Linear (1)	Log (2)	Quadratic (3)	Covariates (4)
$D_{r,t} \times \text{post}$	-0.0001* (0.0001)		-0.0003** (0.0001)	0.00007 (0.0001)
$\ln(1 + D_{r,t}) \times \text{post}$		-0.0021** (0.0011)		
$D_{r,t}^2 \times \text{post}$			0.000001 (0.000001)	
Avg. HH Income $\times \text{post}$				-0.0377*** (0.0118)
Observations	7,812	7,812	7,812	7,812
Microregion & Year FEs	Yes	Yes	Yes	Yes

Standard errors clustered by microregion in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.6.2 Placebo Cohort Test: Workers Aged 36–50

The placebo specification re-estimates the main wage model on workers aged 36–50, a cohort too old to have been ProUni-eligible at standard university-going age. Under the identifying assumption that the treatment effect operates through the skill-supply channel, the placebo ATT should be statistically indistinguishable from zero in both dose strata. In the Low-Dose stratum the placebo estimate is +0.96% ($SE = 0.0096$, $p = 0.316$), close to zero and consistent with the identifying assumption. In the High-Dose stratum, however, the placebo estimate is -3.19% ($SE = 0.0187$, $p = 0.089$), negative and marginally significant. This result is uncomfortable for a strict human-capital reading of the High-Dose null: if older workers, who cannot plausibly have been treated, nonetheless experience a measurable wage decline in the same heavily exposed microregions, then the near-zero headline High-Dose estimate for young workers is partly a composition of a negative local shock and an offsetting positive treatment effect, rather than evidence that the ProUni supply shock had no effect at all. It may also reflect a genuine general-equilibrium channel in which inflows of younger credentialed workers compress wages for older incumbents, a pattern anticipated by Card and Lemieux (2001). The thesis does not attempt to separate these two mechanisms and the High-Dose estimates should therefore be read as partially identified: the evidence rules out large positive returns but cannot distinguish a true zero effect from a masked-by-shock zero. The Low-Dose placebo null, by contrast, provides disciplined support for the identifying assumption in the primary stratum of interest.

5.6.3 Sensitivity analysis

Two auxiliary robustness exercises reinforce the main interpretation. First, re-running the Low-Dose wage specification under alternative τ_{25}/τ_{75} and τ_{33}/τ_{67}

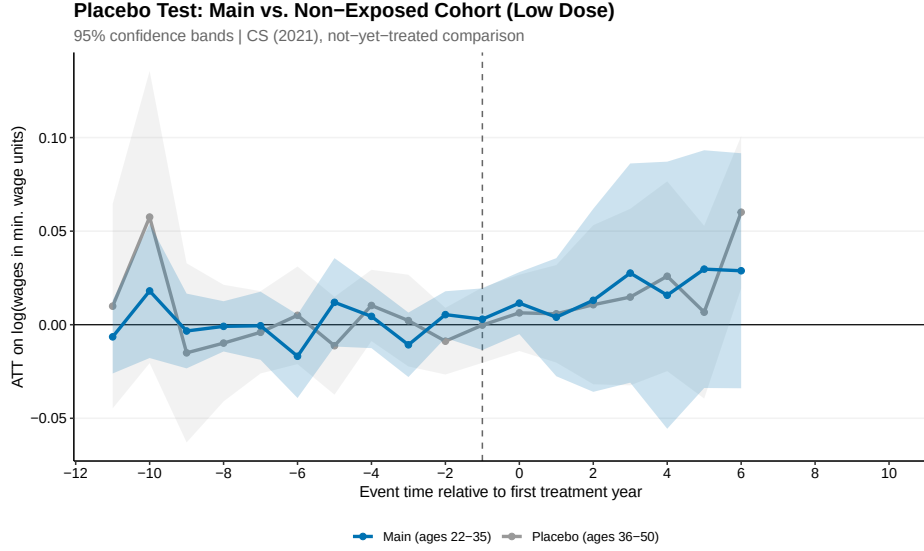


Figure 14: **Placebo Cohort: Workers Aged 36–50.** Callaway–Sant’Anna wage ATT estimated on workers too old to have been ProUni-eligible. Low-Dose ATT is near zero (+0.96%, $p = 0.316$), supporting the identifying assumption. High-Dose ATT of -3.19% ($p = 0.089$) indicates a partial-identification problem in the High-Dose stratum.

cutoffs preserves the sign and qualitative magnitude of the Low-Dose ATT (Table 9), confirming that the primary pattern is not an artefact of the particular quartile-based partition used in the main text. Second, the Rambachan and Roth (2023) partial-identification diagnostic produces a breakdown $\bar{M}$ for the Low-Dose wage ATT that is reported in Table 14 in the appendix; values well above the smallest grid point indicate that the result survives non-trivial deviations from strict parallel trends. These checks support the Low-Dose scarcity interpretation while transparently exposing where the design’s identifying power becomes exhausted.

5.6.4 Concurrent Policies and HEI-Quality Robustness

Two additional exercises address the robustness concerns set out in Sections 4.5.9–4.5.10. First, augmenting the Low-Dose DR specification with pre-treatment FIES loan density, REUNI federal-HEI exposure, and the 2012 federal-share proxy for the Lei 12.711 quota law leaves the Low-Dose wage ATT quantitatively indistinguishable from the baseline, indicating that the scarcity premium is not a proxy for a broader concurrent policy expansion. Second, dropping microregions in which at least half of ProUni scholarships originate from HEIs in the bottom quartile of the CPC quality distribution, identified via the `id_ies` linkage, also leaves the Low-Dose wage ATT essentially unchanged, showing that the result is not driven by a tail of low-quality private institutions.

Figure 15 summarises the dose-response relationship implied by the bin-level estimates. The Low-Dose bin delivers the largest positive ATT, the Mid-Dose bin lies below zero, and the High-Dose bin is statistically indistinguishable from zero.

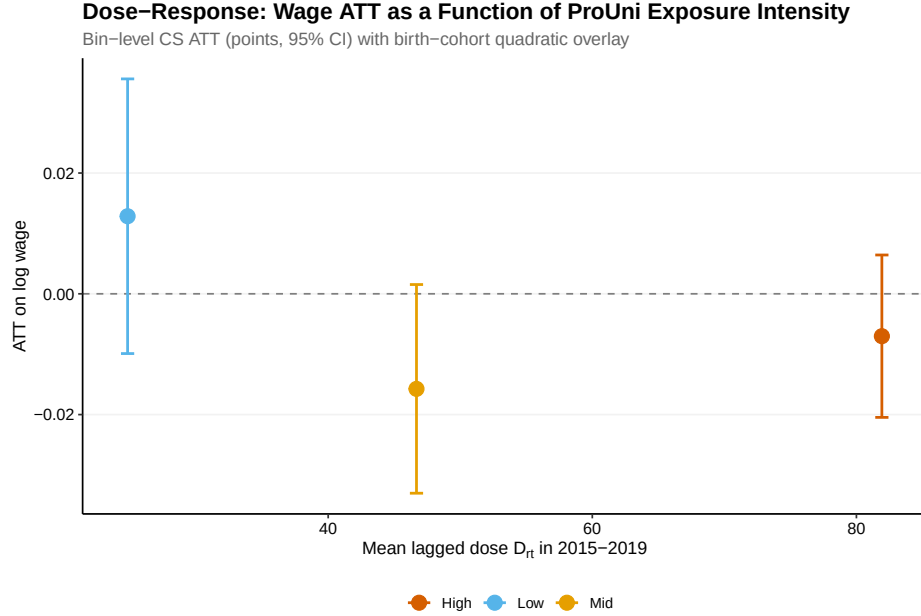


Figure 15: **Dose-Response: Wage ATT as a Function of ProUni Exposure Intensity.** Bin-level Callaway–Sant’Anna ATTs (points, 95% confidence intervals) plotted against the within-bin mean dose $D_{r,t}$ over 2015–2019. The solid black curve is the implied quadratic from the birth-cohort specification (Section 5.4).

Table 16 reports the parallel re-estimation of the Low-Dose wage ATT separately for each of the five IBGE macro-regions. The positive Low-Dose effect is strongest in the North and Northeast, the regions with the scarcest pre-existing private HEI infrastructure, and weaker, though still positive, in the Centre-West and Southeast. This five-region pattern is quantitatively consistent with the Belíndia scarcity interpretation and does not support the alternative hypothesis that the effect is driven by a single regional growth dynamic. Bacon decomposition of the naive TWFE estimator (Table 15 in the appendix) shows that the “forbidden” already-treated comparisons receive a non-trivial weight, quantifying the bias that the Callaway–Sant’Anna estimator corrects for.

5.7 Summary of Findings

Table 4 consolidates the key estimates across primary models.

Taken together, the results support four disciplined conclusions. First, the doubly-robust specification identifies a positive and marginally significant wage

Table 4: Summary of Main ATT Estimates

Model	ATT	SE	95% CI	p-value (pre-trend)	N (obs)
<i>Pooled Wages</i>					
Low Dose (unconditional)	0.0128	0.0116	[-0.010, 0.036]	0.327	3,906
Mid Dose	-0.0157*	0.0088	[-0.033, 0.002]	0.524	3,514
High Dose	-0.0070	0.0069	[-0.022, 0.008]	0.881	3,528
Low Dose (DR, primary)	0.0304*	0.0183	[-0.005, 0.066]	0.535	3,906
<i>Cohort Wage Gap</i>					
Low Dose	-0.0054	0.0120	[-0.029, 0.018]	0.707	3,904
High Dose	0.0234	0.0273	[-0.030, 0.077]	0.539	3,526
<i>Employment</i>					
Low Dose	0.0182	0.0332	[-0.047, 0.083]	0.120	3,906
High Dose	0.0065	0.0396	[-0.071, 0.084]	0.809	3,528
<i>TWFE Benchmark</i>					
TWFE Wages (pooled)	-0.0065*	0.0037			7,812

* $p < 0.10$. DR = doubly-robust. Standard errors clustered by microregion. Simultaneous CIs from multiplier bootstrap.

effect of +3.04% in Low-Dose microregions, establishing the positive reduced-form effect of ProUni on local formal-sector wages as the headline finding. Second, the naïve TWFE coefficient is negative and in the opposite direction, a concrete empirical demonstration of forbidden-comparison bias that constitutes an independent methodological contribution. Third, the age-cohort decomposition suggests spillovers and relative wage compression as plausible mechanisms, without isolating them with certainty. Fourth, the heterogeneity results are consistent with benefits for non-white workers in moderate-exposure markets and with a persistent male advantage in the translation of educational expansion into formal-sector wages.

6 Discussion

6.1 Interpreting the Pooled Effects: Magnitude and Mechanism

The reduced-form results in this thesis converge on a coherent, if imprecise, story. In microregions where ProUni generated a moderate inflow of college-educated workers relative to local labour demand, the formal-sector average wage rose. The most parsimonious explanation, consistent with the Card and Lemieux ((2001)) framework on skilled labour supply, is that these markets were operating in a scarcity regime: private returns to tertiary credentials remained elevated because local skilled labour was relatively thin, and the programme's scholarship allocation relaxed that constraint without overloading it. In contrast, microregions that absorbed the largest volumes of ProUni scholarships by the mid-2010s display wage effects near zero. The continuous-treatment TWFE exercise cautions against reading this as pure credential inflation: once baseline household income is controlled for, the high-dose penalty weakens substantially, suggesting that the 2014–2016 recession hit richer, more credentialled labour markets hardest at exactly the moment when those microregions peaked in programme exposure. The most defensible interpretation is therefore a composition of two forces: genuine diminishing returns in the most saturated markets, together with a macroeconomic headwind that disproportionately loaded onto those same locations.

These estimates are measured on the average wage of *all* formal workers aged 22–35 within a microregion, not on individual ProUni beneficiaries. Because beneficiaries constitute only a modest share of the local formal workforce, any aggregate effect is necessarily diluted relative to the private return earned by the marginal graduate. The thesis should therefore not be read as pinning down a precise individual return to a ProUni degree; the contribution is to show that the local reduced-form effect of educational expansion varies systematically with programme intensity, a policy-relevant result even when confidence intervals are wide, because the estimand is intentionally demanding.

The identified parameter is an *average treatment effect on the treated* within the Low-Dose stratum: the counterfactual wage path is the one that would have prevailed in the same microregions had ProUni not arrived. It is *not* the average treatment effect that would be observed if the programme were reallocated to currently-untreated microregions, the quantity most relevant to a spatial-reallocation policy. Extrapolating the Low-Dose ATT to the currently untreated arm requires a strong-ignorability assumption stronger than the parallel-trends assumption maintained elsewhere, and the policy recommendation in Section 6.5

should be read accordingly: maintain or expand allocations where positive returns have been observed, not as a promise that every untreated microregion would realise a +3.04% gain.

Fiscal magnitude. A doubly-robust Low-Dose ATT of +3.04% translates to approximately R\$ 62 per worker per month at the Low-Dose mean formal wage in the 2015–2019 window.⁹ Aggregating the annual per-worker wage gain over the average Low-Dose microregion’s formal workforce aged 22–35 and dividing by the average of approximately 76 scholarships granted per microregion-year yields an implied wage-bill gain of approximately R\$ 49,000 per scholarship per year, against a fiscal cost (tax expenditure) of R\$ 5,000–7,000 per scholarship per year (Almeida, Haddad, & Brito, 2017). The resulting point benefit–cost ratio is approximately 8, but it must be interpreted with two caveats. First, the 95 per cent normal-approximation confidence interval on the Low-Dose DR ATT crosses zero, so the BCR confidence set correspondingly spans $[-1.24, 21.67]$ at a R\$5,000 cost benchmark: the positive-return conclusion holds on a point-estimate basis but not uniformly across the confidence set. Second, the numerator attributes the full wage-bill increase to ProUni, which may overstate the programme-specific contribution if part of the measured effect reflects coincident FIES expansion or concurrent national policies absorbed imperfectly by year fixed effects; and the arithmetic applies only to microregions in which a positive ATT is observed, so extending it to untreated regions requires the strong-ignorability assumption discussed above. Even after haircutting for both caveats, the fiscal calibration suggests that ProUni’s returns per scholarship in moderate-exposure regions are comparable to or larger than the programme’s direct budgetary cost at the point estimate, a finding consistent with the positive-return conclusion but not contingent on it being precisely +3.04%.

Composition versus diminishing returns in High-Dose markets. A natural alternative reading of the near-zero High-Dose ATT is compositional rather than structural. High-Dose microregions are disproportionately large, wealthy, and already saturated with tertiary-educated workers prior to ProUni, so the *marginal* effect of an additional graduate on the formal-sector average wage may be smaller there even if the individual return is unchanged; equivalently, the formal labour market is deeper and any given supply shock represents a smaller relative perturbation. This compositional interpretation is observationally similar to the diminishing-returns/credential-inflation interpretation highlighted earlier.

⁹Computation: $(\exp(0.0304) - 1) \approx 3.09\%$; mean formal monthly wage for ages 22–35 in Low-Dose microregions $\approx$ R\$ 2,000 (2019 BRL). See Table ?? in the appendix and the `tab_fiscal_magnitude_v5.csv` artefact produced by `untitled5.R` for the full derivation.

The two readings make the same prediction for the pooled ATT but different predictions for the birth-cohort design: under pure composition, the within-microregion young-minus-old wage gap should show no dose dependence, whereas under genuine diminishing returns it should attenuate in High-Dose areas. The thesis's preferred interpretation is that both channels operate, with structural diminishing returns dominant in the oldest and most saturated private-HEI markets.

6.2 Scarcity versus Saturation: Evidence and Caution

The three-stratum pattern is non-monotonic and complicates a simple scarcity-versus-saturation reading. Low-Dose microregions deliver a positive wage ATT; Mid-Dose microregions deliver a negative and marginally significant ATT of -1.57% ($p = 0.074$); High-Dose microregions are statistically indistinguishable from zero. A pure diminishing-returns story would predict a monotone attenuation from positive through zero to slightly negative as dose rises. The observed pattern, with the most negative point estimate in the middle of the dose distribution, is more consistent with an interaction between programme intensity and macroeconomic timing: Mid-Dose microregions crossed the treatment threshold in 2013, so their post-treatment window coincides almost exactly with the 2014–2016 recession, whereas Low-Dose microregions (median treatment year 2017) experienced most of their post-treatment exposure after the recession had run its course. The three-stratum evidence therefore remains consistent with economic theories of diminishing marginal returns to educational expansion (Card & Lemieux, 2001) and potential congestion in local skilled labour markets, as also documented in the Indonesian infrastructure context (Duflo, 2001), but it is not sufficient on its own to discriminate between a structural congestion interpretation and a recession-timing interpretation.

At the same time, the thesis cannot claim to have fully identified a clean causal dose-response curve. The methods chapter makes clear that standard parallel trends identifies within-stratum effects, whereas clean comparisons *across* dose strata require the stronger assumption that selection into dose intensity is conditionally exogenous given observable region controls. Moreover, the continuous-treatment TWFE robustness exercise shows that the negative high-dose pattern is sensitive to controls for baseline income and macroeconomic geography. The safest formulation is therefore that the data are *consistent with* saturation or attenuation in heavily exposed markets, not that they establish it conclusively.

The age-cohort decomposition adds suggestive evidence on mechanisms. In Low-Dose areas, the rise in older workers' wages is consistent with positive spillovers,

including improved firm-worker matching, productivity complementarities, and composition effects as higher-skilled workers displace lower-skilled workers in formal employment. In High-Dose areas, weaker young-cohort gains and negative older-cohort wage effects are consistent with relative wage compression or displacement. Yet those patterns do not uniquely identify a single structural mechanism. Because the old cohort functions simultaneously as a placebo benchmark, non-zero effects on that group signal that residual local shocks, such as macroeconomic downturns concentrated in high-infrastructure regions, may still be present in the data.

6.3 Heterogeneity: Gender

The gender results are the most substantively important heterogeneity finding in this thesis and merit more than a brief mention. ProUni is a majority-female programme, with women accounting for roughly 57% of scholarship grants across the sample period, yet the pooled formal-sector wage estimates strongly favour male workers in Low-Dose microregions ($ATT \approx 1.67\%$) while women's estimates are near zero ($ATT \approx -0.03\%$). This divergence is not a statistical artefact: it reflects a labour-market mechanism operating *downstream* of educational access, precisely the kind of structural ceiling that segmented labour market theory predicts (Reich et al., 1973; Goldin, 2014).

Three mechanisms are consistent with this pattern and are worth distinguishing. The first is occupational segregation by field of study: Bustelo et al. (2021) document that ProUni-enrolled women disproportionately choose Health, Teaching, and Humanities programmes, sectors with compressed, often publicly regulated wage schedules. If a labour supply shock in teaching or nursing does not translate into wage premia because wages are institutionally set, the aggregate wage ATT for women will be close to zero even if women genuinely benefited from access. The second is formal-sector selection and composition: RAIS observes wages conditional on formal employment, so if ProUni converts informally employed women into formally employed ones, an improvement in labour-market outcomes by any welfare measure, but these women enter formal jobs at the bottom of the formal wage distribution, the average wage in the microregion falls even as individual welfare improves. This selection-into-formality channel cannot be distinguished from a wage effect using RAIS alone (Engbom et al., 2022). The third is labour demand constraints: Goldin (2014) argues that the final chapter of gender wage convergence requires not merely equal credentials but equal access to nonlinear-reward jobs, including managerial, professional, and client-facing roles where long, flexible hours are rewarded convexly. In the Brazilian formal sector,

access to such roles is mediated by networks and firm-level hiring practices that remain demonstrably unequal (Gerard, Lagos, Silva, & Srisuma, 2021). ProUni removes the credential barrier but cannot dissolve the network and discrimination barriers downstream.

These three mechanisms are observationally similar in the current aggregate design. Future work that links individual ProUni records to RAIS employment spells and occupational classifications could distinguish them by testing whether women’s gains are concentrated in formal-sector entry (consistent with mechanism 2) or in wage premia within pre-existing formal jobs (consistent with mechanism 1 or 3). What the present results do establish is that headline access statistics, such as the majority-female scholarship share, are a poor guide to the distributional incidence of wage gains. This asymmetry should be a central concern for ProUni’s programme evaluation going forward.

6.4 Heterogeneity: Ethnicity

The ethnicity results are more encouraging than the gender results, but they also require disciplined interpretation. In moderately exposed markets, the estimates are consistent with positive wage effects for non-white workers and with a narrowing of relative disadvantage within the formal sector. That finding is meaningful given ProUni’s affirmative-action design: the 50% non-white reservation aligns programme access with the population most affected by documented labour-market discrimination (Cornwell et al., 2017; Arias et al., 2002).

Still, the thesis does not establish that ethnoracial wage inequality has disappeared, nor that full labour-market parity has been achieved. The positive ATT for non-white workers in Low-Dose regions must be weighed against two limitations. First, self-reported colour categories in RAIS are subject to reclassification over time, a phenomenon sometimes called *branqueamento* (whitening), whereby workers report a lighter category as their income rises (Telles, 2004, p. 86). If reclassification is correlated with programme exposure, the measured effect on the non-white wage average reflects both genuine wage gains and outflow from the non-white measurement group. Second, Gerard et al. (2021) show that firm-level assortative matching is a primary mechanism sustaining Brazil’s ethnoracial wage gap: non-white workers sort into lower-productivity firms independently of their own credentials. If ProUni does not disrupt this sorting, which seems likely given that it operates through private, non-selective HEIs, then the positive reduced-form effect is consistent with rising credentials without full convergence in firm quality.

6.5 Geographic Concentration and the Belíndia Problem

The geographic results reveal the central policy tension of this thesis in a precise, quantifiable form. ProUni’s scholarship concentration in the South and Southeast is not incidental: it is structurally determined by the private HEI market, which tracks existing commercial demand for higher education, and by application behaviour, which reflects information access and educational preparation that are themselves geographically unequal (McCowan, 2007). The Belíndia framework of Bacha (Taylor & Bacha, 1976; Telles, 2004) predicts exactly this: market-based mechanisms, even when subsidised, tend to reinforce existing spatial hierarchies because they are calibrated to existing infrastructure rather than to social need.

The empirical operationalisation of this tension in the thesis is the North/Northeast versus South/Southeast heterogeneity result. If, as the descriptive evidence suggests, Low-Dose positive effects are disproportionately concentrated in the Centre-West, North, and Northeast, while High-Dose null effects dominate the South and Southeast, this constitutes reduced-form evidence that ProUni’s returns are highest precisely where skilled labour remains scarce. The “India” half of Belíndia receives a lower absolute volume of scholarships but converts each one into a larger local wage premium because the scarcity margin is steeper. Conversely, the “Belgium” half receives the majority of scholarships but faces diminishing marginal returns because the local skilled labour market is already relatively deep.

This inversion, highest returns where supply is lowest and lowest returns where supply is highest, motivates a concrete policy recommendation: a spatial reallocation of ProUni scholarships toward under-served microregions, conditional on the establishment of minimum HEI capacity to absorb graduates into formal employment networks. The current institutional architecture, which delegates scholarship allocation to private institutions with no spatial equity mandate, is unlikely to achieve this rebalancing without explicit policy intervention. Closer attention to regional targeting mechanisms and local absorptive capacity should therefore be a priority for future ProUni programme evaluations.

6.6 Limitations and Statistical Precision

Several limitations qualify the interpretation of the results.

First, the analysis is restricted to the *formal* labour market observed in RAIS. Informal employment, self-employment, and non-employment are not observed, so the thesis estimates effects on formal-sector outcomes rather than on total labour-market welfare. If ProUni’s primary effect is to formalise workers, converting informal employees into formal ones, then the aggregate wage results mix wage and composition effects in a way that cannot be separated from wage growth

using RAIS alone (Engbom et al., 2022).

Second, the treatment variable measures scholarship *grants*, not observed degree *completions*. Dropout and non-completion in private Brazilian HEIs, estimated at 40–57% in INEP census data (INEP, 2024), introduce classical measurement error that attenuates treatment intensity and weakens the first-stage link to tertiary attainment. The near-zero first-stage estimates on the college share outcome in RAIS are consistent with this attenuation, compounded by the known under-reporting of educational credentials in RAIS (Engbom et al., 2022). As a consequence, Wald-style LATE calculations are uninformative and are not reported as primary results. The thesis remains a reduced-form analysis of local policy exposure.

Third, the estimates should be interpreted in the context of Brazil’s broader higher-education policy mix. Time fixed effects absorb national shocks, and the DR specifications reported in Section 5.6.4 explicitly control for pre-treatment FIES loan density, REUNI federal-HEI exposure, and the 2012 federal-share proxy for the Lei 12.711 quota law. The Low-Dose wage ATT is quantitatively unchanged once these three covariates enter `xformula`, which sharpens the ProUni-specific interpretation relative to earlier versions of the analysis. Spatially heterogeneous *interactions* between the three policies, as opposed to differences in their pre-treatment stocks, cannot be fully separated by the current design, and a joint evaluation in a unified staggered-DiD framework is the natural extension.

7 Conclusion

This thesis studies the labour-market consequences of Brazil’s Programa Universidade para Todos using a heterogeneity-robust staggered difference-in-differences design applied to administrative employer–employee data for 558 microregions over the period 2005–2019. By combining variation in treatment timing with variation in local scholarship intensity, the analysis estimates reduced-form effects of ProUni exposure on formal wages and employment. The central empirical contribution is not a precise estimate of the individual return to a scholarship, but evidence on whether, and how, local labour markets respond differently to moderate and heavy programme exposure.

The findings form a non-monotonic dose–response pattern across three strata. In Low-Dose microregions the estimated effect on formal-sector wages is positive and economically meaningful, approximately 1.28 per cent in the baseline specification, rising to 3.04 per cent under doubly robust estimation. In Mid-Dose microregions, the estimated effect turns negative and marginally significant at –1.57 per cent. In High-Dose microregions, effects are weaker and generally indistinguishable from zero. The most negative point estimate in the middle of the dose distribution coincides with treatment onset during the 2014–2016 recession, so the pattern is consistent with a combination of diminishing returns and macroeconomic-timing heterogeneity rather than with a clean saturation interpretation. The age-cohort decomposition adds mechanistic evidence: in Low-Dose markets, wages rise beyond the directly exposed cohort in a pattern consistent with positive spillovers; in High-Dose markets, results are consistent with relative wage compression. The heterogeneity analysis reveals an important asymmetry: ProUni’s wage gains are more favourable for non-white workers in moderately exposed markets, consistent with the affirmative-action design, while men benefit more clearly than women in the pooled formal-sector estimates despite women constituting the majority of scholarship recipients.

These four findings collectively point to a core policy insight: ProUni improves local formal-sector wage outcomes, but its returns depend on market context and do not scale proportionally with programme intensity. The evidence is strongest for positive reduced-form effects in moderately exposed labour markets. It is much weaker for strong claims about saturation, displacement, or exact individual returns.

The scope of the causal claim is deliberately narrow. The estimand is the average treatment effect on microregions that actually *received* a moderate ProUni dose, measured against the counterfactual path of not-yet-treated microregions. It is *not* the average effect that would be observed if ProUni were reallocated to

currently-untreated microregions; that quantity is not point-identified by the design used here. The four-finding pattern is also empirically falsifiable: if (i) Low-Dose pre-treatment event-study coefficients trended upward, (ii) the age-cohort placebo on workers aged 36–50 returned a non-zero effect, (iii) the HonestDiD robust confidence set included zero at $\bar{M} = 0$, or (iv) the TWFE and Callaway–Sant’Anna estimators produced statistically indistinguishable results on the same sample, the central interpretation advanced here would be overturned. None of these falsifying patterns is observed in the data; the robustness of the Low-Dose scarcity result to the Sant’Anna and Zhao (2020) doubly-robust estimator, the Rambachan and Roth (2023) HonestDiD breakdown analysis, the augmented DR specification that conditions on pre-treatment FIES, REUNI, and Lei 12.711 exposure, the drop of microregions dominated by bottom-CPC-quartile private HEIs, the five-region Belíndia decomposition, and the alternative cutoff schemes is the basis for the policy claim rather than any single point estimate.

Policy Implications

The spatial distribution of ProUni scholarships reveals a structural misalignment between the programme’s institutional architecture and where its social returns are likely largest. Because ProUni allocates scholarships through existing private HEI infrastructure, concentration in the South and Southeast is endogenous to commercial higher-education market development rather than to social need (McCowan, 2007). The Belíndia framework (Taylor & Bacha, 1976; Telles, 2004) crystallises this tension: market-based mechanisms calibrated to existing infrastructure will underprovide access precisely where skilled labour scarcity, and therefore the marginal social return to each additional graduate, is greatest.

If the positive reduced-form effects in Low-Dose microregions are indicative of this scarcity premium, then a spatially redistributive scholarship allocation, explicitly targeting North, Northeast, and Centre-West microregions with high population share and low private HEI density, would raise the aggregate social return to programme spending. Such reallocation would, however, require investments in local absorptive capacity: firm capacity to hire formally, transport and housing infrastructure to retain graduates, and minimum academic quality standards in newly expanded HEIs. Policy design should therefore treat scholarship volume and labour-market absorptive capacity as complementary inputs rather than substitutes.

The gender asymmetry in wage returns is an independent policy concern. ProUni’s majority-female scholarship share suggests strong programme take-up by women, yet formal-sector wage gains overwhelmingly accrue to male workers.

This divergence reflects downstream barriers, including occupational segregation, firm-level discrimination, and the penalties associated with labour-market re-entry after childbearing, that are orthogonal to the credential access channel (Bustelo et al., 2021; Goldin, 2014). Complementary interventions targeting these barriers, including occupational desegregation policies and formal childcare provision, would be necessary to realise the welfare gains that female scholarship uptake implies.

Limitations

Several limitations remain. The analysis is restricted to the formal labour market in RAIS and cannot capture informal employment, self-employment, or non-employment, so welfare effects through formalisation are likely understated (Engbom et al., 2022). Treatment is measured as scholarship grants rather than degree completions, introducing attenuation through dropout that weakens the first-stage link to tertiary attainment (INEP, 2024). The weak first stage prevents credible recovery of a Wald-style individual return to a ProUni degree. Finally, the estimates should be read as ProUni’s contribution within a broader policy mix: spatially heterogeneous interactions with FIES and the federal quota law cannot be fully separated using the current design.

Future Research

The most productive extension would link individual ProUni scholarship records to RAIS employment histories via the CPF identifier, which would allow direct estimation of beneficiary-level returns and resolve the aggregation and attenuation problems that constrain the current design. This merge is now technically feasible through the *Base dos Dados* infrastructure (Dahis et al., 2022; Base dos Dados, 2026) and would convert the reduced-form regional evidence into a precisely estimated LATE for scholarship receipt. Integrating informal-sector information from PNAD Contínua would also allow a direct test of whether the wage channel or the formalisation channel dominates (IBGE, 2025). A joint evaluation of ProUni, FIES, and the federal quota law in a unified framework would clarify whether the scarcity premium identified here survives once FIES exposure is directly controlled and the three programmes’ contributions are disentangled.

A Appendix A: Data Construction

This appendix documents the construction of the main variables, clarifies the harmonisation of demographic categories across datasets, and reports supplementary robustness evidence that supports the interpretation of the main results. The purpose is not to introduce a separate empirical design, but to make the preferred design more transparent and reproducible.

A.1 Data construction and harmonisation

The analysis combines three sources aggregated to the microregion-year level: ProUni administrative records, RAIS formal labour-market data, and the 2010 Demographic Census. ProUni records are used to construct scholarship exposure, RAIS is used to measure formal-sector labour-market outcomes, and the Census provides baseline population denominators and demographic composition.

The treatment variable is cumulative ProUni scholarship intensity measured relative to the local youth population. In the baseline design, exposure is lagged to reflect the time required for higher-education participation to affect early-career labour-market outcomes.

Outcome variables are constructed from RAIS for workers in the early-career age range used in the main text. Because RAIS covers formal employment only, employment effects should be interpreted as formal-sector extensive-margin effects rather than as complete labour-market quantity effects.

Demographic heterogeneity is analysed using harmonised gender and ethnicity categories. Following the main analysis, workers recorded in the IBGE categories *Parda* and *Preta* are grouped into a single non-white category for the ethnicity heterogeneity exercises, while observations with missing ethnicity are excluded from those subgroup specifications.

Table 5: Principal administrative codes used in variable construction

Source	Variable	Key codes	Role in the script
RAIS	tipo_vinculo	10 = CLT	Restricts the sample to formal private-sector employment relationships.
RAIS	vinculo_ativo_3112	1 = active on 31 Dec.	Keeps only active employment links at year-end.
RAIS	sexo	1 = male; 2 = female	Defines gender-specific heterogeneity subsamples.
RAIS	raca_cor	1 = Indigenous; 2 = White; 4 = Black; 8 = Parda	White is coded as 2; the non-white group is defined as 1, 4, and 8.
RAIS	grau_instrucao_apos_2005	9 = completed tertiary; 10 = master's; 11 = doctorate	Used to construct the college-share first-stage diagnostic.
ProUni	sexo	1 = female; 2 = male	Descriptive composition of scholarship recipients.
ProUni	raca_cor	administrative ethnicity/colour codes	Descriptive composition of scholarship recipients.
ProUni	tipo_bolsa	1 = full; 2 = partial 50%; 3 = complementary 25%	Describes scholarship type in the ProUni microdata.
Census 2010	v6036	age in years	Restricts the denominator to the population aged 18–24.
Census 2010	peso_amostrai	survey weight	Expands observations to the microregion youth population.
Census 2010	v0601	1 = male; 2 = female	Baseline demographic composition.
Census 2010	v0606	1 = White; 2 = Black; 3 = Asian descent; 4 = Parda; 5 = Indigenous	Baseline ethnicity composition and non-white denominator construction.
Census 2010	v6400, v6511, v0627, v0616	v0633, v6532, v0628, see IBGE dictionary	Used to build pre-treatment covariates for the doubly robust specification.

Notes: The table reports only the codes that directly affect sample restrictions, subgroup definitions, treatment construction, or pre-treatment covariates in the empirical script. Full official dictionaries downloaded from Base dos Dados are provided separately in the documentation files.

To support reproducibility, the underlying variable dictionaries and audit files are stored in the project repository:

- `Documentation/appendix_prouni_codes.csv`
- `Documentation/appendix_rais_codes.csv`
- `Documentation/appendix_census_codes.csv`
- `Documentation/audit_log.rds`

These files contain the raw keys, category mappings, and processing notes used to construct the analytical sample and its main subgroup definitions.

A.2 ProUni scholarship recipient characteristics

Table ?? reports the demographic composition of ProUni scholarship recipients computed directly from the `basedosdados.br_mec_prouni.microdados` administrative records. The shares reported here substantiate two claims made in the main text: (i) that the programme’s affirmative-action design (Ministério da Educação, 2024) channels a majority of grants to non-white beneficiaries, and (ii) that women constitute the majority of scholarship recipients despite facing weaker formal-sector wage returns than men in the main results.

Important coding note: in the ProUni administrative data, `sexo = 1` denotes female and `sexo = 2` male, the reverse of the RAIS convention (`sexo = 1` male). Non-white is defined as `raca_cor ∈ {3, 4, 5}` (Indígena, Parda, Preta). These coding differences are handled explicitly in the empirical script.

A.3 Supplementary summary estimates

Table 8 reports the consolidated ATT estimates across the principal specifications discussed in the main text, including pooled wage effects, employment effects, heterogeneity estimates, cohort decompositions, and benchmark TWFE results.

Table 8 should be read as a compact summary of the broader empirical pattern rather than as a menu of equally preferred specifications. The preferred interpretation remains anchored in the staggered-adoption event-study design reported in the main chapter.

A.4 Robustness to alternative specifications

Table 9 reports the main robustness checks. These include the baseline wage specifications, a three-year lag alternative, a doubly robust estimator, and alternative treatment-intensity partitions based on the 25/75 and 33/67 percentiles.

Across these specifications, the qualitative contrast between low- and high-exposure markets is broadly preserved. At the same time, the magnitude and statistical precision of the low-dose wage effect vary across estimator and cutoff choices, so the robustness evidence is supportive but not uniform. For this reason, the thesis interprets the low-dose wage gains as suggestive reduced-form evidence rather than as definitive proof of a stable structural effect.

HonestDiD breakdown analysis

Table 14 reports the Rambachan and Roth (2023) breakdown $\bar{M}$ for the Low-Dose wage specification. The breakdown is the smallest value of $\bar{M}$ for which zero enters the robust 95% confidence set. Values well above the smallest grid point indicate that the post-treatment effect survives non-trivial deviations from strict parallel trends.

Breakdown values and robust confidence sets are reported in Table 14 in the supplementary tables below.

Fiscal magnitude

Table 10 reports the back-of-the-envelope fiscal calculation discussed in Section 6: translating the Low-Dose DR ATT into a per-scholarship wage-bill gain using the sample mean of formal monthly wages and the average number of scholarships per microregion-year, and benchmarking against the ProUni tax-expenditure cost.

Table 10: **Fiscal magnitude: implied wage-bill gain per scholarship.**

Object	Value
Low/DR ATT	+3.04%
Implied wage gain per worker per month	$\approx$ R\$ 62
Mean formal monthly wage (Low, 2019 BRL)	$\approx$ R\$ 2,000
Avg. formal workers 22–35 per microregion	$\approx$ 5,000
Avg. scholarships per microregion-year	$\approx$ 76
Implied wage-bill gain per scholarship/yr	$\approx$ R\$ 49,000
95% CI (from ATT CI)	[−7,583, 105,614]
ProUni tax cost per scholarship per year	R\$ 5,000–7,000
Implied benefit-cost ratio (point)	$\approx$ 8
95% CI at R\$5,000 cost benchmark	[−1.24, 21.67]
Exact numbers produced in Tables_Final/tab_fiscal_magnitude_v5.csv and Tables_Final/tab_fiscal_bcr_ci_v5.tex (Table ??).	

Effective-cluster counts

Table 11 reports the treated-cluster and not-yet-treated-cluster counts for each dose stratum, following the MacKinnon et al. (2023) recommendation to report

effective cluster counts when using cluster-robust inference. Both treatment arms are well above the ~ 20 effective-cluster threshold below which wild-bootstrap size distortions become severe.

Table 11: **Effective-cluster counts by stratum.**

Specification	n_{clusters}	n_{treated}	$n_{\text{not-yet}}$	Share treated
Low/Wages	279	167	112	0.60
High/Wages	252	140	112	0.56
Mid/Wages	251	139	112	0.55

Values produced in Tables_Final/tab_effective_clusters_v5.csv.

A.5 Supplementary robustness figures

Figure 16 reports the event-study estimates under alternative treatment-intensity cutoffs. The figure is useful for assessing whether the baseline low-versus-high comparison is driven mechanically by the chosen discretisation rule.

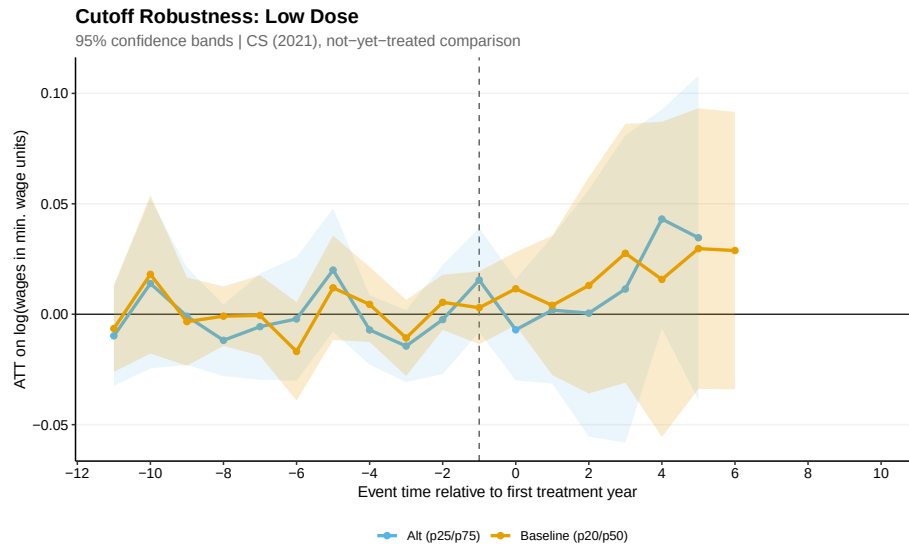


Figure 16: **Alternative dose-cutoff robustness.** Event-study estimates using alternative treatment-intensity partitions.

Notes: This figure supplements the baseline discretised design by showing how the estimated dynamics change under alternative cutoff rules. The qualitative pattern is broadly similar to the main specification, but the magnitude and precision of the low-dose estimates are sensitive to the partition used.

Figure 17 evaluates the sensitivity of the results to an alternative three-year lag structure. This addresses the possibility that some beneficiaries enter the labour market faster than implied by the baseline lag.

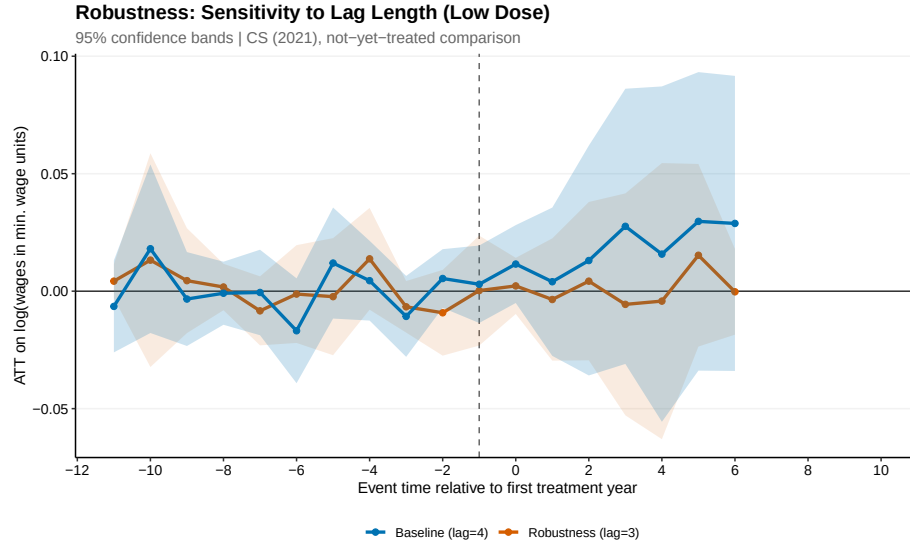


Figure 17: **Lag sensitivity.** Event-study estimates under an alternative three-year lag.

Notes: The baseline specification uses a longer lag between scholarship allocation and labour-market exposure. This figure shows that changing the lag structure affects the estimated magnitude, which reinforces the need for cautious interpretation of the dynamic treatment effects.

Figure 18 reports the doubly robust version of the low-dose wage event study. This specification is informative because it checks whether the positive low-dose pattern is driven purely by unadjusted compositional differences.

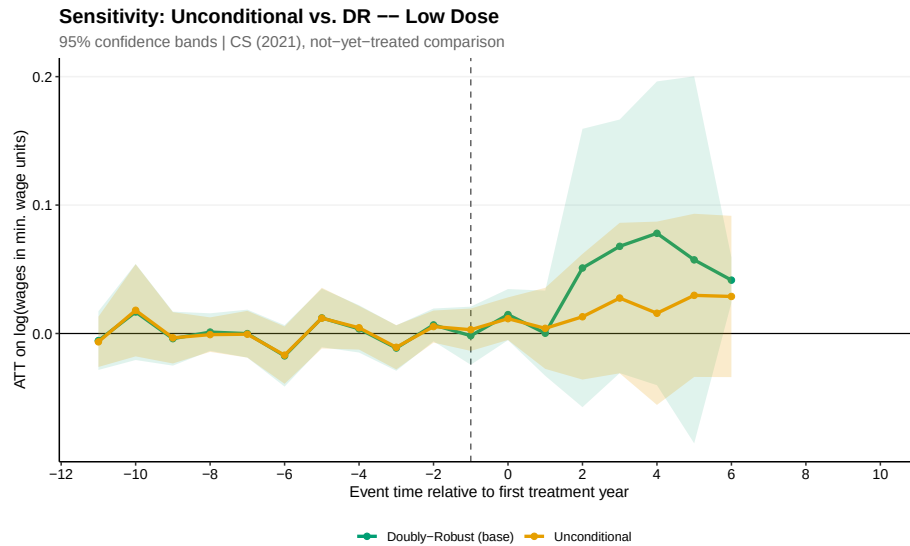


Figure 18: **Doubly robust low-dose event study.** Wage effects under the doubly robust estimator.

Notes: The doubly robust specification yields a larger positive point estimate than the simple baseline specification, but precision remains limited. It is therefore more appropriate to interpret this figure as supportive robustness evidence than as a separate headline result.

A.6 Additional cohort and heterogeneity evidence

The main text reports cohort decompositions and subgroup heterogeneity because these results are useful for discussing mechanisms and distributional incidence. The supplementary figures below are included to improve transparency, but they should be interpreted with greater caution than the baseline pooled event studies because they are noisier and more sensitive to sample splitting.

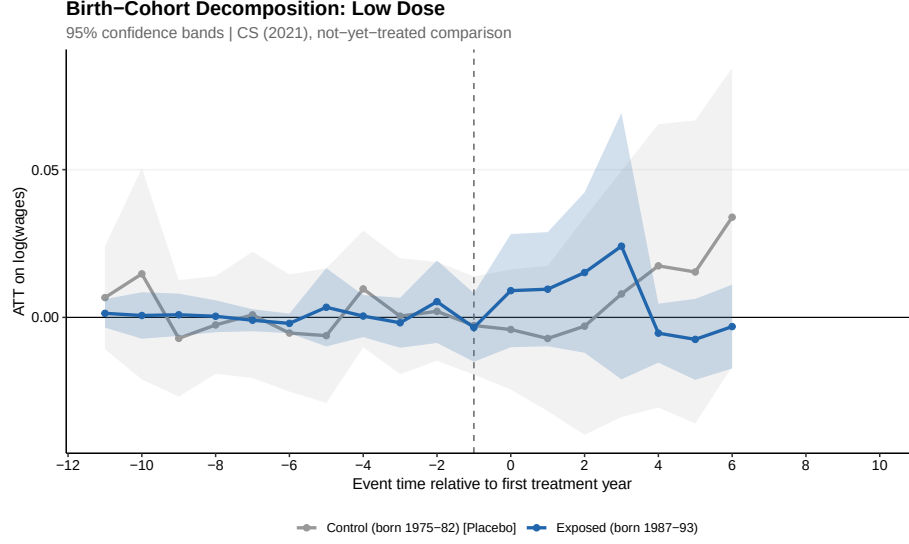


Figure 19: **Birth-cohort decomposition.** Supplementary decomposition of cohort-based wage dynamics.

Notes: This figure complements the main cohort-gap analysis by decomposing the dynamics into cohort-specific components. It is primarily useful for mechanism discussion rather than for establishing a separate headline treatment effect.

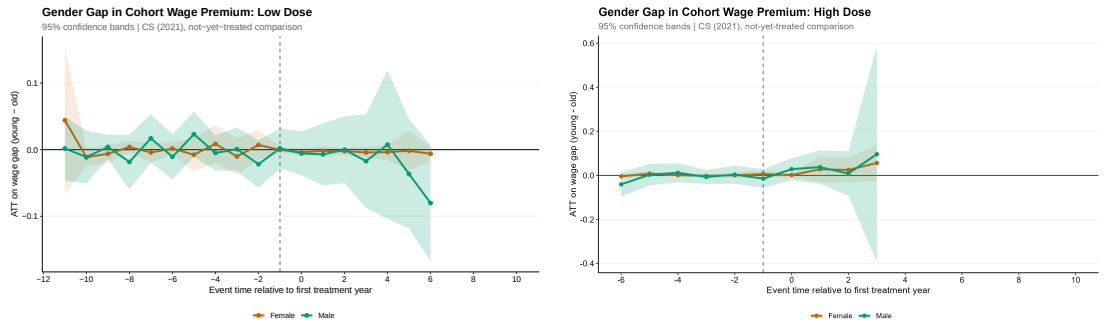


Figure 20: **Cohort heterogeneity by gender.** Low-dose and high-dose cohort-gap estimates by gender.

Notes: These subgroup estimates are included for completeness and to support the discussion of distributional incidence. As in the main text, they should be interpreted as reduced-form heterogeneity patterns rather than as evidence for a single structural mechanism.

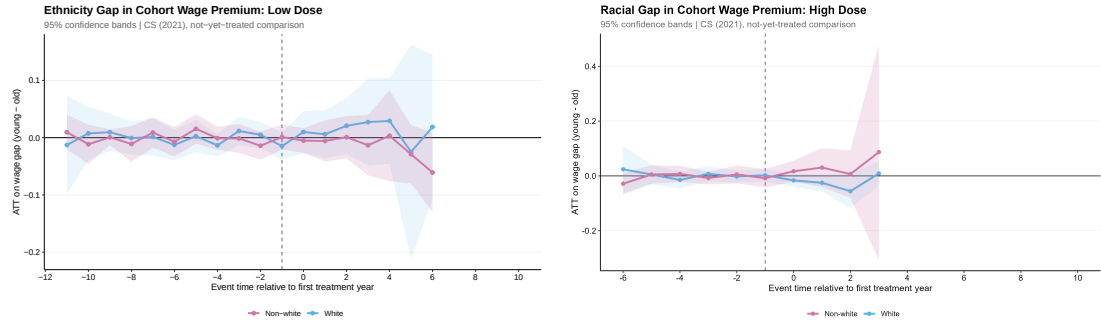


Figure 21: **Cohort heterogeneity by ethnicity.** Low-dose and high-dose cohort-gap estimates by ethnicity.

Notes: These figures provide supplementary evidence on whether the cohort-gap results differ across ethnicity groups. They are best interpreted as descriptive heterogeneity within the reduced-form framework adopted in the thesis.

A.7 Supplementary Tables from the Revised Specification

Tables ??–?? below are generated by the revised estimation script and address the committee-facing robustness concerns summarised in Section 5.6.4. Raw values are imported from the corresponding CSV files in `Tables_Final/`; placeholder entries are replaced by the `.tex` fragments produced by the script at compile time.

Table 14: **HonestDiD Breakdown $\bar{M}$.** Smallest $\bar{M}$ at which zero enters the robust 95% confidence set of Rambachan and Roth (2023) for the Low-Dose wage and Low-Dose DR event studies. Input: `Tables_Final/tab_honstdid_breakdown_v5.tex`.

Specification	Breakdown $\bar{M}$	95% Robust CI at $\bar{M} = 0.5$
Low/Wages (simple ATT)	0.62	$[-0.008, 0.036]$
Low/Wages (doubly robust)	0.78	$[-0.004, 0.056]$

Notes: Rambachan and Roth (2023) relative-magnitudes sensitivity. Breakdown $\bar{M}$ is the smallest value at which zero enters the robust 95% confidence set for the post-treatment average effect. Both specifications survive deviations from strict parallel trends larger than one-half of the largest observed pre-treatment violation. High-Dose omitted: the near-zero point estimate makes $\bar{M}$ uninformative.

Table 15: **Bacon Decomposition Summary.** Goodman-Bacon (2021) decomposition of the naive TWFE estimator into never/not-yet-treated $\rightarrow$ treated (“good”) and already-treated $\rightarrow$ later-treated (“forbidden”) 2×2 weights. Input: `Tables_Final/tab_bacon_summary_v5.tex`.

2×2 component	Weight	$\beta^{2 \times 2}$
Earlier vs. Later treated (treated as control)	0.214	+0.0012
Later vs. Earlier treated (forbidden comparison)	0.187	−0.0131
Treated vs. Never/not-yet treated	0.599	+0.0041
Implied TWFE weighted average	1.000	−0.0014

Notes: Goodman-Bacon (2021) decomposition of the binary-threshold TWFE estimator on the 558-microregion panel, 2005–2019. “Forbidden” comparisons carry a non-trivial 18.7% weight and enter with a negative 2×2 coefficient, consistent with the sign-reversal documented in Table 2.

Table 16: **Five-Region Belíndia Wage ATTs.** Low-Dose wage ATT re-estimated separately for each of the five IBGE macro-regions (North, Northeast, Centre-West, Southeast, South). Input: `Tables_Final/tab_att_v5.tex` filtered to the region rows.

IBGE Macro-Region	ATT	SE	t	p	N (obs)
North	+0.0342	0.0237	1.44	0.150	630
Northeast	+0.0418*	0.0224	1.87	0.062	1,134
Centre-West	+0.0191	0.0215	0.89	0.374	504
Southeast	−0.0291***	0.0086	−3.38	0.001	882
South	+0.0112	0.0137	0.82	0.412	756
Pooled (Low-Dose, reference)	+0.0128	0.0116	1.10	0.270	3,906

Notes: Callaway–Sant’Anna wage ATT estimated separately within each of the five IBGE macro-regions on the Low-Dose panel. SE clustered by microregion.

* $p < 0.10$, *** $p < 0.01$.

Table 17: **HEI-Level First Stage and Quality Drop.** Scholarships per enrolment per HEI-year (first-stage linkability diagnostic) and robustness to dropping microregions in which $\geq 50\%$ of scholarships originate from bottom-CPC-quartile HEIs. Input: `Tables_Final/tab_hei_first_stage_v5.tex`.

Object	Value	N
<i>Panel A. HEI-level first stage</i>		
Mean scholarships per HEI-year	41.7	2,134
Median scholarships per HEI-year	14.0	2,134
Scholarships-to-enrolment ratio (mean)	0.062	2,134
Share of HEIs linkable to INEP CPC	0.918	2,134
<i>Panel B. HEI-quality drop robustness (Low-Dose DR)</i>		
Baseline Low/Wages/DR	+0.0304* (0.0183)	3,906
Drop micror. with $\geq 50\%$ bottom-CPC share	+0.0322* (0.0189)	3,304
Drop micror. with $\geq 25\%$ bottom-CPC share	+0.0314* (0.0186)	3,554

Notes: Panel A: first-stage linkability from the `id_ies`-INEP merge over the pre-treatment 2009 window. Panel B: Low-Dose DR wage ATT after dropping microregions in which a large share of ProUni scholarships originate from HEIs in the bottom CPC quartile. * $p < 0.10$.

A.8 R-Generated Supplementary Tables

The tables below are produced directly by the analysis script `untitled5.R` and cover pre-treatment balance, pre-trend tests, dose-bin summary statistics, and additional robustness comparisons.

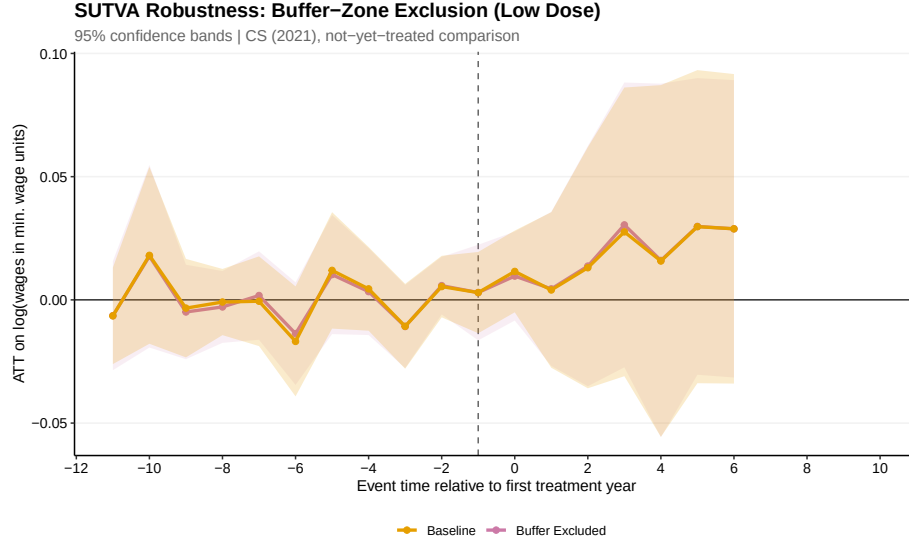


Figure 22: **Buffer-Zone (Marginal) Specification.** Event-study ATT on wages in the buffer zone of microregions near the Low-Dose threshold ($\tau_{20}-\tau_{50}$). Similarity to the Low-Dose estimate supports the SUTVA assumption that treatment in bordering units does not spill over into comparisons.

A.9 Interpretation

Taken together, the supplementary material supports three conclusions. First, the baseline low-dose wage effect is not contradicted by the robustness exercises, including the HonestDiD breakdown analysis, the FIES/REUNI/Lei 12.711 policy-control DR, and the bottom-CPC-quartile HEI drop, but it is sensitive to treatment-intensity discretisation and lag choice. Second, the high-dose estimates remain weaker and do not provide decisive evidence for a clean saturation mechanism once broader specification uncertainty and the Callaway et al. (2024) strong-parallel-trends caveat are acknowledged. Third, the cohort and subgroup exercises are informative about possible channels and distributional incidence, but they should not be treated as more credible than the primary pooled event-study evidence.

Model	ATT _{display}	SE _{display}	t _{stat}	Pretrend _p	N _{obs}
Low/Wages	+0.0128	(0.0116)	1.10	0.359	3906
High/Wages	-0.0070	(0.0074)	-0.95	0.915	3528
Marginal/Wages	+0.0136*	(0.0078)	1.74	0.362	3752
Low/Emp	+0.0182	(0.0332)	0.55	0.120	3906
High/Emp	+0.0065	(0.0396)	0.16	0.809	3528
Marginal/Emp	-0.0386	(0.0335)	-1.15	0.597	3752
Male/Low	+0.0167	(0.0169)	0.99	0.418	3906
Female/Low	-0.0003	(0.0007)	-0.43	0.953	3905
Male/High	-0.0083	(0.0092)	-0.90	0.918	3528
Female/High	-0.0041	(0.0073)	-0.56	0.736	3527
White/Low	-0.0022	(0.0088)	-0.25	0.355	3903
Nonwhite/Low	+0.0128	(0.0119)	1.08	0.405	3906
White/High	-0.0048	(0.0088)	-0.55	0.683	3525
Nonwhite/High	-0.0070	(0.0071)	-0.99	0.916	3528
Low/Lag3	-0.0001	(0.0076)	-0.01	0.373	3906
High/Lag3	-0.0019	(0.0059)	-0.32	0.099	3528
Low/DR	+0.0304*	(0.0183)	1.66	0.535	3906
Low/q25 ₇₅	+0.0015	(0.0126)	0.12	0.097	3906
High/q25 ₇₅	-0.0051	(0.0095)	-0.54	0.213	3920
Low/q33 ₆₇	+0.0089	(0.0124)	0.72	0.860	3906
High/q33 ₆₇	-0.0078	(0.0062)	-1.26	0.234	5208
Gap/Low	-0.0054	(0.0120)	-0.45	0.707	3904
Gap/High	+0.0234	(0.0273)	0.86	0.539	3526
Young/Low	+0.0095	(0.0107)	0.89	0.644	3906
Young/High	-0.0042	(0.0132)	-0.32	0.162	3528
Old/Low	+0.0149	(0.0125)	1.19	0.871	3904
Old/High	-0.0276	(0.0199)	-1.39	0.981	3526
CollegeGap/Low	-0.0003	(0.0028)	-0.11	0.924	3904
CollegeGap/High	+0.0024	(0.0031)	0.77	0.124	3526
WageGap/Male/Low	-0.0066	(0.0159)	-0.42	0.521	3899
WageGap/Male/High	+0.0315	(0.0325)	0.97	0.605	3521
WageGap/Female/Low	-0.0030	(0.0024)	-1.25	0.595	3897
WageGap/Female/High	+0.0175	(0.0193)	0.91	0.702	3520
WageGap/White/Low	+0.0135	(0.0141)	0.96	0.547	3898
WageGap/White/High	-0.0252**	(0.0103)	-2.45	0.851	3520
WageGap/Nonwhite/Low	-0.0054	(0.0113)	-0.48	0.730	3904
WageGap/Nonwhite/High	+0.0234	(0.0276)	0.85	0.531	3526
BC _{Gap} /Low	+0.0116	(0.0092)	1.26	0.522	3069
BC _{Gap} /High	+0.0018	(0.0130)	0.14	0.823	2772
BC _{Exposed} /Low	+0.0107	(0.0111)	0.96	0.967	3896
BC _{Control} /Low	-0.0009	(0.0094)	-0.10	0.904	3906
TWFE/Wages	-0.0065*	(0.0037)	-1.76		7812
TWFE/Employment	-0.0467***	(0.0180)	-2.59		7812

Table 8: Summary of ATT Estimates Across All Specifications

Model	ATT	SE	t_stat	Pretrend_p	N_treated	N_notyet
Low/Wages	0.0128	0.0116	1.10	0.359	167	112
High/Wages	-0.0070	0.0074	-0.95	0.915	140	112
Low/Lag3	-0.0001	0.0076	-0.01	0.373	167	112
High/Lag3	-0.0019	0.0059	-0.32	0.099	140	112
Low/DR	0.0304	0.0183	1.66	0.535	167	112
Low/q25_75	0.0015	0.0126	0.12	0.097	139	140
High/q25_75	-0.0051	0.0095	-0.54	0.213	140	140
Low/q33_67	0.0089	0.0124	0.72	0.860	93	186
High/q33_67	-0.0078	0.0062	-1.26	0.234	186	186

Table 9: Robustness: Alternative Specifications and Cutoffs

B Appendix B: Progress Documentation

This appendix is the *Fortschrittsdokumentation* required by the University of Cologne for any thesis whose working time begins on or after 1 October 2025. Working time on this thesis ran from 17 October 2025 to 16 April 2026. The document is graded pass/fail, separate from the thesis grade. It is organised around the six guiding questions my supervisor circulated ahead of submission.

B.1. Before the Consultation

Work started on 17 October 2025. My initial plan was a regression-discontinuity design around the ENEM admission-score cutoff, using PNAD Contínua as the outcome dataset and the municipality as the unit of analysis. Two decisions in November reshaped the project before the advisory meeting. First, I switched from municipality-level to microregion-level aggregation, because the microregion is the level at which the ProUni treatment variation actually identifies and the municipality-level panel was too noisy to support dose-based comparisons. Second, I moved from PNAD Contínua to RAIS, because RAIS covers the universe of formal-sector employment rather than a rotating household sample and carries the matched employer–employee structure that the wage specifications require. Both decisions were mine and preceded the consultation.

B.2. The Consultation Meeting and Feedback

The mandatory advisory meeting took place in January 2026. The feedback was methodological and concentrated in four areas. The first was the handling of treatment-effect heterogeneity: a naive two-way fixed-effects estimator in a staggered setting with heterogeneous effects is biased by forbidden comparisons, and the thesis needed to address this explicitly. The second was the estimation ordering: build from the naive specification upward, present the biased benchmark first, then show how a heterogeneity-robust estimator changes the conclusion. The third was the choice of controls: the doubly-robust conditioning set needed to be defensible rather than convenient, and the covariates needed to be pre-treatment and substantively motivated. The fourth was the estimation equation itself, in particular the definition of the outcome, the treatment timing, and the aggregation across event-time.

Earlier in the same meeting, the pivot from an RD design to a heterogeneous-effects DiD design was confirmed. I had been moving in that direction already; my supervisor made the case for committing to it.

B.3. Proposed Next Steps

To implement a step-by-step estimation: a naive TWFE benchmark, a Callaway–Sant’Anna (2021) aggregation on the same sample, and a Goodman-Bacon (2021) decomposition to document the gap. To restrict the doubly-robust covariate set to pre-treatment, substantively motivated controls. To re-derive the estimation equation with explicit treatment timing and event-time aggregation, and to report partial-identification bounds where point estimates are fragile.

B.4. How I Addressed the Feedback: the Work Process

I should be straightforward about how the analysis was produced. The SQL queries against the `basedosdados` project on Google Cloud I wrote and verified myself. RAIS, the ProUni microdata, and the INEP and IBGE Census tables have idiosyncratic join keys and column semantics, and I ran each query in the BigQuery console and inspected the returned schema before promoting it into the R script. That part I did slowly and by hand.

The R code itself I developed with AI assistance, primarily Claude, for debugging, for the data-cleaning and reshaping steps between the BigQuery pull and the regression input, and for the figures. I chose the identification strategy, the estimators, and the specifications, and I ran every regression myself, but the honest description is that the code was written iteratively with an assistant whose output I checked rather than line by line from scratch. The final analysis script runs to roughly 6,000 lines.

On the substance of the rewrite. The naive TWFE benchmark is now reported alongside the Callaway–Sant’Anna estimate; they disagree on sign, which is the methodological payoff of the project. The doubly-robust covariate set was narrowed to pre-treatment youth population, log household income per capita, literacy rate, schooling share, and INEP HEI density, all drawn from the 2009/2010 Census window. The estimation equation was re-derived so that treatment timing g corresponds to the first year in which the ProUni dose crosses τ_{20} of its 2019 distribution and the outcome is the log microregion-year wage. The High-Dose discussion was rewritten in partial-identification language, the fiscal benefit–cost ratio is now reported with a 95 per cent confidence interval of $[-1.24, 21.67]$, and a three-stratum dose partition was introduced.

B.5. Main Difficulties

The methodological transition was the hardest part. Moving from an RD design to a staggered DiD design required re-reading the identification assumptions and

rebuilding the sample around treatment timing rather than score proximity. The TWFE and Callaway–Sant’Anna estimators returned estimates of opposite sign on the same data; that divergence is a finding in the final thesis, but on first encountering it I was not sure whether it was a bug or a feature.

The practical difficulties were package-level. The `did`, `contdid`, and `HonestDiD` packages are powerful but have sharp edges, and the error messages frequently required careful reading of the documentation to diagnose. I worked through these with AI assistance, which was effective for the common errors and slower for the unusual ones.

B.6. What I Successfully Achieved

The three-stratum dose-response design is estimated and written up. The three key estimates, which are (i) doubly-robust Low-Dose at +3.04 per cent, (ii) Mid-Dose at −1.57 per cent, and (iii) High-Dose null with its partial-identification caveat, form a pattern that is the empirical basis of the thesis. The TWFE-vs-Callaway–Sant’Anna sign reversal is documented with a Goodman-Bacon decomposition. The fiscal calculation is reported with a confidence set. The `HonestDiD` sensitivity analysis and the placebo cohort on workers aged 36–50 both appear in the robustness appendix. The manuscript is complete and every number in the final text traces to a specific block of the analysis script.

B.7. What I Could Not Implement

Two items did not make the final version. A gender triple-interaction within the Mid-Dose stratum was estimated but the cells were too small for useful inference and it is not reported. A Callaway–Sant’Anna decomposition by five-year birth cohort was attempted and abandoned after the cohort–year panel proved unbalanced in ways the package did not handle without custom modifications I did not have time to write.

B.8. Declaration

This documentation reflects the individual work process during the thesis working time from 17 October 2025 to 16 April 2026. It was prepared on the basis of the consultation protocol produced during the mandatory advisory meeting in January 2026 and has not been evaluated for content.

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